ITA0329-MOBILE COMPUTING
ASSIGNMENT 3

Name: Duddela Harika
Reg no: 192324301

ASSIGNMENT 3 – CO3

CO3 Coverage: Introduction to wireless networks – mobile IP - issues in mobile IPdifferences between cellular and ad hoc wireless networks- applications - routing protocols – Proactive & reactive Routing Protocols.

Mode: Pair Work – 2 Students per Group

Duration: 1 Week

Total Marks: 50

Pair-Work Principle

Both students are responsible for understanding the complete assignment.

However, each question contains clearly identified individual responsibilities so that the contribution and understanding of each student can be independently verified.

Student A and Student B must not divide the assignment question-wise. Both students must participate in every question.

The individual responsibilities intentionally alternate across questions so that both students demonstrate evidence of:

Concept Understanding → Network Analysis → Protocol Design → Implementation/Simulation → Evaluation

Q1 – Wireless Network Architecture and Mobility Analysis 6 Marks

Part A – Introduction to Wireless Networks

Shared Group Work

Together:

- Understand the characteristics and components of wireless networks.
- Identify mobile nodes, wireless access points/base stations and communication links.
- Develop a suitable wireless network architecture.
- Identify bandwidth, connectivity and mobility constraints.
- analyse communication between mobile users and network infrastructure.
- Implement or simulate a small wireless communication scenario and test the communication flow.

Student A – Network Architecture Analysis

Student A must individually:

- Draw and represent the wireless network architecture.

- Identify the major network components and their roles.
- analyse wireless communication characteristics.
- Analyse the effect of limited bandwidth and wireless connectivity.
- analyse how mobile nodes communicate through the wireless infrastructure.

Student B – Mobility and Performance Analysis

Student B must individually:

- Design representative mobility scenarios.
- Analyse handoff and connectivity interruption.
- Study packet loss and transmission delay.
- Analyse the effect of changing network conditions.
- Compare communication behaviour under different mobility conditions.

Part B – Wireless Mobility Scenario

Shared Group Work

Together, develop and analyse a scenario in which a mobile user moves between wireless coverage areas.

Student A – Analytical Investigation

Student A must:

- Define successful communication and interrupted communication cases.
- Identify the sequence of mobility events.
- Record communication delay and packet delivery behaviour.
- Investigate increasing mobility conditions.
- Present the analytical evidence.

Student B – Mobility Logic

Student B must:

- Formulate the mobility decision sequence.
- analyse when a mobile node changes its point of attachment.
- Identify which communication conditions are affected.
- analyse the termination or completion condition for the scenario.
- Relate repeated mobility to communication efficiency.

Individual Verification

Faculty may ask Student A to analyse the wireless architecture and Student B to analyse the mobility scenario. Therefore, primary responsibility does not mean exclusive knowledge.

Q2 – Mobile IP and Mobility Management 7 Marks

Shared Group Work

The pair must:

- Study Mobile IP architecture and terminology.
- Identify Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- analyse agent discovery and registration.
- Analyse packet delivery to a mobile node.
- analyse tunnelling and mobility support.
- Conduct a small simulation or worked example of Mobile IP communication.

Student A – Mobile IP Operation

Student A is individually responsible for:

- analyseing the role of the Mobile Node.
- analyseing the role of the Home Agent.
- analyseing the role of the Foreign Agent.
- analyseing the role of the Correspondent Node.
- Describing registration and location updating.
- analyseing the packet-delivery and tunnelling process.

Student B – Mobile IP Challenge Analysis

Student B is individually responsible for:

- Identifying Mobile IP challenges.
- Analysing registration delay.
- Analysing tunnelling overhead.
- analyseing triangular routing.
- Analysing packet loss during mobility.
- interpreting security and scalability considerations.

Joint Design Question

Together answer:

If a mobile user frequently changes its network of attachment during an active communication session, what problems can occur in Mobile IP and how can they be reduced?

Evidence Expected

- Mobile IP architecture diagram.
- Registration/packet-flow sequence.
- Identification of mobility-related delays and overhead.
- Reasoned justification of the proposed improvement.

Cross-Understanding Requirement

Student A must be able to analyse the major Mobile IP challenges.

Student B must be able to analyse the complete Mobile IP registration and packet delivery process.

Both students must be able to defend the answer.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks 7 Marks

Shared Group Work

Together:

- Study cellular wireless network architecture.
- Study ad hoc wireless network architecture.
- Identify infrastructure-based and infrastructure-less communication.
- Compare mobility and topology characteristics.
- Compare routing requirements.
- Develop a technical comparison diagram and table.

Student A – Cellular Network Investigation

Student A must individually:

- Represent cellular network architecture.
- analyse the role of base stations and infrastructure.
- Analyse centralized communication.
- analyse handoff between cells.
- Analyse scalability and coverage.
- Identify suitable cellular applications.

Student B – Ad Hoc Network Investigation

Student B must individually:

- Represent ad hoc network architecture.
- analyse communication between mobile nodes.
- Analyse dynamic topology.
- analyse multi-hop communication.
- Analyse the absence of fixed infrastructure.
- Identify suitable ad hoc applications.

Comparative Analysis

Parameter	Cellular Network	Ad Hoc Network
Infrastructure		
Communication		
Routing		

Parameter	Cellular Network	Ad Hoc Network
Mobility		
Topology		
Scalability		
Handoff		
Failure Handling		
Applications		

Joint Engineering Decision

Together determine:

For an emergency/disaster environment where fixed infrastructure may be unavailable, which network architecture would you select and why?

Cross-Defence

Student A must analyse why cellular networks depend on infrastructure.

Student B must analyse how ad hoc networks continue communication without fixed infrastructure.

Q4 – Applications of Mobile IP and Routing Requirements 7 Marks

Shared Group Work

Together:

- Identify real-world applications requiring mobility support.
- Select one application scenario such as emergency communication, mobile healthcare or intelligent transportation.
- Identify the mobile entities and communication requirements.
- Apply Mobile IP concepts to the selected scenario.
- Identify the routing requirements created by mobility.
- Evaluate the suitability of the proposed solution.

Student B – Application Architecture

Student B must individually:

- Select and describe an appropriate Mobile IP application.
- Identify mobile nodes and communication endpoints.
- Design the application network architecture.
- analyse how Mobile IP supports user mobility.
- Identify required network components.
- analyse the communication flow.

Student A – Performance and Requirement Analysis

Student A must individually:

- Analyse communication delay.
- Analyse packet delivery and connectivity.
- Identify mobility requirements.
- Analyse interruption during movement.
- Identify security and reliability requirements.
- Evaluate Mobile IP suitability.

Q5 – Proactive Routing Protocols in Mobile Networks 6 Marks

Shared Group Work

Together:

- Study proactive routing protocols.
- Understand continuous routing-table maintenance.
- Select a suitable protocol such as DSDV.
- Develop a mobile-node topology.
- Analyse routing-table updates.
- Evaluate routing behaviour under mobility.

Student A – Protocol Design

Student A must individually:

- analyse the principle of proactive routing.
- analyse DSDV operation.
- analyse routing-table maintenance.
- analyse sequence numbers and route updates.
- Draw a sample routing topology.
- Demonstrate route information before and after a topology change.

Student B – Performance Investigation

Student B must individually:

- Analyse routing-control overhead.
- Analyse route availability.
- Study convergence behaviour.
- Analyse control-message transmission.
- Compare behaviour for different network sizes.
- Evaluate delay and packet delivery.

Q6 – Reactive Routing and Protocol Comparison 7 Marks

Student A – Reactive Routing Investigation

Student A takes primary responsibility for reactive routing.

- analyse the concept of reactive routing.
- Study AODV.
- analyse Route Request (RREQ).
- analyse Route Reply (RREP).
- analyse Route Error (RERR).
- Analyse route discovery and route maintenance.
- analyse the effect of mobility on route discovery.

Student B – Comparative Verification

Student B must provide a comparative analysis of AODV and DSDV.

- Compare route discovery mechanisms.
- Compare routing-table requirements.
- Analyse control overhead.
- Compare initial communication delay.
- Analyse mobility handling.
- Identify suitable application scenarios.
- Justify the selection of a protocol for a given network condition.

Joint Question

For a highly dynamic mobile network with frequent topology changes, would you select DSDV or AODV? Justify your decision using routing overhead, delay and mobility considerations.

DAY-4 DESIGN CHALLENGE

Q7 – Design a Mobile Emergency Communication Network 10 Marks

The Day-4 problem must be solved collaboratively.

Neither student may work on this question independently and simply combine the answers later.

Scenario

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move continuously across different geographical areas. Fixed communication infrastructure may become unavailable, while mobile users must exchange critical information with minimum delay.

Stage 1 – Independent Design Proposal

Before interpreting the solution with each other:

Student A

Independently propose:

- A suitable wireless network architecture.
- A mobility-management strategy.
- The role of Mobile IP.
- A possible communication model.

Student B

Independently propose:

- A suitable routing strategy.
- Proactive/reactive protocol selection.
- Route maintenance approach.
- Failure-handling mechanism.
- Performance parameters.

Each student's initial proposal must be retained in the submission.

Stage 2 – Compare the Two Designs

The pair must compare where the designs agree, where they differ, strengths and weaknesses, Mobile IP requirements, routing suitability, expected overhead, expected delay and handling of node mobility.

Stage 3 – Joint Final Design

Using the strongest ideas from the interpretion, jointly develop the final mobile emergency communication solution.

- Network architecture showing mobile nodes, wireless links and routing infrastructure.
- Mobile IP components: Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- Registration and tunnelling process.
- Selected routing protocol and justification.
- Failure handling for node/link failure and network disconnection.
- Performance evaluation using delay, packet delivery and routing overhead.

Stage 4 – Divide the Validation Responsibilities

Student A – Analytical Validation

- Analyse Mobile IP overhead.
- Analyse mobility-related delay.
- Evaluate packet delivery.
- Analyse communication interruption.
- Justify the selected mobility-management mechanism.

Student B – Experimental Validation

- Design network test cases.
- Test different numbers of mobile nodes.
- Test different mobility conditions.
- Compare proactive and reactive routing behaviour.
- Record delay and packet-delivery results.
- Present experimental evidence.

Stage 5 – Cross-Defence

Student A must analyse: Why was the selected routing protocol appropriate for the emergency mobile network?

Student B must analyse: How does Mobile IP maintain communication when a mobile node changes its network location?

Stage 6 – Joint Evaluation

- Why the proposed network architecture is suitable.
- Why Mobile IP is required or not required.
- Why the selected routing protocol is appropriate.
- What network overhead is introduced.
- How mobility affects performance.
- What happens when connectivity is lost.
- Under what conditions the proposed solution may not be appropriate.

Conclude: What characteristics should a network designer analyse before selecting Mobile IP and a routing protocol for a highly mobile wireless environment?

CONTRIBUTION RECORD

Question	Student A – Primary Contribution	Student B – Primary Contribution	Joint Work
Q1	Network Architecture	Mobility Analysis	Wireless Network Analysis
Q2	Mobile IP Operation	Mobile IP Challenges	Mobile IP Architecture
Q3	Cellular Network	Ad Hoc Network	Comparative Analysis
Q4	Performance Analysis	Application Architecture	Application Evaluation
Q5	Protocol Design	Performance Investigation	Proactive Routing
Q6	Reactive Routing	DSDV–AODV Comparison	Routing Evaluation
Q7	Analytical Validation	Experimental Validation	Final Network Design

INDIVIDUAL VERIFICATION

Faculty may select any question, including a component for which the student was not designated as the primary contributor.

- analyse wireless network architecture.
- analyse Mobile IP operation and registration.
- analyse mobility challenges.
- Compare cellular and ad hoc wireless networks.
- analyse applications of Mobile IP.
- analyse proactive routing.
- analyse reactive routing.
- Compare DSDV and AODV.
- Justify protocol selection.
- Interpret experimental results.
- Justify network design decisions.
- Respond to a modified network condition.

A student should not answer: “My partner did this section.”

Understanding of the complete submitted work is a requirement of the assignment.

NOTE: RECOMMENDED FACULTY VERIFICATION MODEL (VIVA)

For each student, conduct a brief 4–5 minute verification.

Question 1 – Primary responsibility

Ask one technical question from the component for which the student had primary responsibility.

Question 2 – Partner's responsibility

Ask one question from a component primarily completed by the partner.

Question 3 – Transfer question

Modify a condition and ask the student to reason about the consequence.

- If a mobile node moves from one network to another during an active session, what happens to its IP connectivity and how does Mobile IP address the problem?
- If the number of mobile nodes increases significantly, how would AODV and DSDV differ in routing overhead?
- If the network topology changes very frequently, which routing approach would you select and why?
- If the fixed infrastructure becomes unavailable during a disaster, how would an ad hoc network support communication?

This verification establishes individual attainment without requiring a second complete individual assignment.

SIMATS ENGINEERING
MOBILE COMPUTING
STANDARD FOR ALGORITHMIC PSEUDOCODE

Applicable to: Mobile Computing Assignments and Exercises

1. What is Mobile Computing Pseudocode?

Mobile Computing pseudocode is a clear, precise and language-independent representation of a mobile networking algorithm. It describes mobility management, packet forwarding, route discovery, route maintenance, handoff, and routing decisions without depending on Java, Python, C, C++, Android, or simulator-specific syntax.

A good Mobile Computing algorithm must make the network state and communication logic visible. It should show what happens when a mobile node moves, when a route exists or fails, when a registration is required, and when a packet must be forwarded through a tunnel or discovered route.

Remember

- Pseudocode is not English prose.
- Pseudocode is not programming code.
- Pseudocode is an implementation-independent specification of mobile-network logic.
- Every important networking decision must be explicit.
- Abstraction is permitted; ambiguity is not.

2. Standard Structure

Use the following structure for a complete Mobile Computing algorithm.

ALGORITHM AlgorithmName(parameters)
INPUT:
Clearly describe nodes, packets, addresses, routes or network state
OUTPUT:
Clearly describe the expected mobility/routing result
BEGIN
INITIALISE required variables/data structures
Detect or receive relevant network information
Execute mobility/routing decisions
UPDATE network state
RETURN result
END

Example

ALGORITHM CheckMobileNodeLocation(MN, HomeNetwork, CurrentNetwork)
INPUT:
MN – mobile node
HomeNetwork – node's registered home network

CurrentNetwork – network where MN is currently attached
OUTPUT:
Mobility status
BEGIN
IF CurrentNetwork = HomeNetwork THEN
status ← HOME_NETWORK
ELSE
status ← VISITED_NETWORK
END IF
RETURN status
END

3. Standard Keywords and Notation

PURPOSE	USE
Declare algorithm	ALGORITHM
Specify input	INPUT:
Specify output	OUTPUT:
Start algorithm	BEGIN
End algorithm	END
Assignment	←
Condition	IF ... THEN
Alternative	ELSE
Multiple conditions	ELSE IF
Loop	FOR ... DO / WHILE ... DO
Return result	RETURN
Boolean operations	AND, OR, NOT
Packet operation	FORWARD / DELIVER / DROP
Route operation	ADD ROUTE / REMOVE ROUTE / UPDATE ROUTE
Message operation	SEND / RECEIVE
Timer	START TIMER / EXPIRE TIMER

Use explicit state names such as HomeAddress, CareOfAddress, NextHop, SequenceNumber, HopCount, RouteState, LinkState and RegistrationStatus.

4. Wireless Network Representation

Represent a mobile network as a graph G = (V, E), where V is the set of mobile nodes and infrastructure nodes, and E represents currently usable communication links.

NETWORK G = (V, E)
V = {mobile nodes, base stations, access points, gateways}
E = {wireless links currently available}
link(u,v) = communication link between u and v

cost(u,v) = communication cost of link (u,v)
status(u,v) ∈ {UP, DOWN}

Network State

STATE:
Location[MN] = current network/location of mobile node MN
NextHop[D] = next hop toward destination D
RouteState[D] ∈ {VALID, INVALID, UNKNOWN}
SequenceNumber[D] = latest known destination sequence number
HopCount[D] = number of hops in current route
LastUpdate[D] = time of most recent route information

5. Wireless Network Discovery

Wireless nodes must determine whether they are connected to a suitable network and whether neighboring nodes are reachable. The discovery procedure is especially important in dynamic environments.

ALGORITHM DiscoverWirelessNeighbors(MN, TransmissionRange)
INPUT:
MN – mobile node
TransmissionRange – maximum discovery range
OUTPUT:
NeighborSet
BEGIN
NeighborSet ← EMPTY_SET
BROADCAST HELLO from MN
WAIT for neighbor responses
FOR each Response received DO
IF SignalStrength(Response) > MIN_SIGNAL THEN
IF Distance(MN, Response.Node) ≤ TransmissionRange THEN
ADD Response.Node to NeighborSet
END IF
END IF
END FOR
RETURN NeighborSet
END

6. Mobile IP – Basic Operation

Mobile IP allows a mobile node to retain a permanent home address while changing its point of network attachment. The algorithm must maintain a binding between the permanent home address and the current care-of address.

Mobile IP Entities

- Mobile Node (MN)
- Home Network

- Home Agent (HA)
- Foreign Network / Visited Network
- Foreign Agent (FA) in traditional Mobile IPv4 terminology
- Care-of Address (CoA)
- Correspondent Node (CN)

```
ALGORITHM MobileIPLocationUpdate(MN, HA, CurrentNetwork)
INPUT:
MN – mobile node
HA – home agent
CurrentNetwork – visited network
OUTPUT:
Registered care-of address
BEGIN
IF CurrentNetwork = MN.HomeNetwork THEN
CoA ← MN.HomeAddress
RegistrationStatus ← HOME
ELSE
CoA ← ObtainCareOfAddress(CurrentNetwork)
SEND RegistrationRequest(MN.HomeAddress, CoA) TO HA
WAIT FOR RegistrationReply
IF RegistrationReply = ACCEPTED THEN
RegistrationStatus ← REGISTERED
ELSE
RegistrationStatus ← FAILED
END IF
END IF
RETURN CoA, RegistrationStatus
END
```

7. Mobile IP Agent Discovery

```
ALGORITHM MobileIPAgentDiscovery(MN)
INPUT:
MN – mobile node
OUTPUT:
Detected mobility agent or network information
BEGIN
LISTEN for AgentAdvertisement
IF Advertisement received THEN
Extract NetworkAddress
Extract AgentType
IF AgentType = HOME_AGENT THEN
Set HomeAgentAddress
ELSE IF AgentType = FOREIGN_AGENT THEN
```

Set ForeignAgentAddress
END IF
RETURN AgentType, NetworkAddress
ELSE
SEND AgentSolicitation
WAIT for Advertisement
IF Advertisement received THEN
RETURN detected agent information
ELSE
RETURN AGENT_NOT_FOUND
END IF
END IF
END

8. Mobile IP Registration

ALGORITHM RegisterMobileNode(MN, HA, CoA)
INPUT:
MN – mobile node
HA – home agent
CoA – current care-of address
OUTPUT:
Registration result
BEGIN
Create RegistrationRequest
Request.HomeAddress ← MN.HomeAddress
Request.CareOfAddress ← CoA
Request.Lifetime ← RequestedLifetime
Request.Identifier ← GenerateRegistrationIdentifier()
Authenticate Request
SEND Request TO HA
WAIT FOR RegistrationReply
IF Reply.Status = ACCEPTED THEN
Binding[MN.HomeAddress] ← CoA
RegistrationExpiry ← CurrentTime + Reply.Lifetime
RETURN REGISTRATION_SUCCESS
ELSE
RETURN REGISTRATION_FAILURE
END IF
END

9. Mobile IP Packet Tunnelling

The home agent can forward packets toward the mobile node's current location by tunnelling them to the registered care-of address.

ALGORITHM TunnelPacketToMobileNode(Packet, DestinationHomeAddress)

INPUT:
Packet – original IP packet
DestinationHomeAddress – mobile node home address
OUTPUT:
Packet delivered through tunnel or DROP
BEGIN
IF Binding[DestinationHomeAddress] does not exist THEN
DROP Packet
RETURN DELIVERY_FAILURE
END IF
CoA ← Binding[DestinationHomeAddress]
EncapsulatedPacket ← ENCAPSULATE(Packet, CoA)
FORWARD EncapsulatedPacket TO CoA
IF TunnelDeliverySuccessful THEN
Decapsulate at tunnel endpoint
DELIVER original Packet TO MobileNode
RETURN DELIVERY_SUCCESS
ELSE
RETURN DELIVERY_FAILURE
END IF
END

10. Mobile IP Challenges

CHALLENGE	ALGORITHMIC IMPLICATION
Handoff delay	Detect movement and update location quickly
Registration overhead	Limit unnecessary signalling
Packet loss	Buffer, retransmit or reroute when appropriate
Triangular routing	Evaluate indirect path and forwarding delay
Security	Authenticate registration and protect mobility messages
Scalability	Avoid excessive control traffic
Wireless errors	Detect link failure and retry or select alternate path
Changing bandwidth	Adapt routing and transmission decisions

Challenge Detection Algorithm

ALGORITHM DetectMobileIPChallenge(MN, Route, Registration)
INPUT:
MN, current route, registration state
OUTPUT:
Identified challenge
BEGIN
IF RegistrationStatus = FAILED THEN
RETURN REGISTRATION_PROBLEM

ELSE IF HandoffInProgress = TRUE THEN
RETURN HANDOFF_DELAY
ELSE IF PacketLossRate > MAX_LOSS THEN
RETURN PACKET_LOSS
ELSE IF RouteDelay > MAX_DELAY THEN
RETURN ROUTING_DELAY
ELSE IF SecurityValidation = FAILED THEN
RETURN SECURITY_PROBLEM
ELSE
RETURN NO_MAJOR_PROBLEM
END IF
END

11. Handoff Management

ALGORITHM PerformHandoff(MN, OldNetwork, NewNetwork)
INPUT:
MN – mobile node
OldNetwork – previous point of attachment
NewNetwork – target network
OUTPUT:
Handoff status
BEGIN
Detect NewNetwork
Measure signal quality
IF SignalQuality(NewNetwork) < MIN_SIGNAL THEN
RETURN HANDOFF_REJECTED
END IF
Obtain new network address or care-of address
Pause non-critical transmissions if required
UPDATE mobility registration
UPDATE routing information
IF RegistrationStatus = SUCCESS THEN
SWITCH active attachment TO NewNetwork
Resume data transmission
RETURN HANDOFF_SUCCESS
ELSE
Restore OldNetwork if available
RETURN HANDOFF_FAILURE
END IF
END

12. Cellular vs Ad Hoc Wireless Networks

PARAMETER	CELLULAR NETWORK	AD HOC NETWORK
Infrastructure	Base stations / access infrastructure	No fixed infrastructure required
Topology	Planned cells	Dynamic topology
Routing	Infrastructure-assisted	Distributed multi-hop routing
Node role	Mobile device mainly communicates through infrastructure	A node can act as host and router
Mobility support	Cellular mobility management	Distributed route discovery/maintenance
Deployment	Requires pre-existing infrastructure	Can be rapidly deployed
Typical environment	Public cellular service	Emergency, military, temporary or field networks
Failure behaviour	Infrastructure failure can affect service	Alternate peer routes may be possible

13. Selecting Cellular or Ad Hoc Architecture

```
ALGORITHM SelectWirelessArchitecture(Scenario)
INPUT:
InfrastructureAvailability
NodeMobility
DeploymentTime
CoverageRequirement
MultiHopRequirement
OUTPUT:
Recommended architecture
BEGIN
IF InfrastructureAvailability = AVAILABLE AND
CoverageRequirement = LARGE THEN
Architecture ← CELLULAR
ELSE IF DeploymentTime = IMMEDIATE AND
InfrastructureAvailability = UNAVAILABLE THEN
Architecture ← AD_HOC
ELSE IF MultiHopRequirement = REQUIRED THEN
Architecture ← AD_HOC
ELSE
Architecture ← CELLULAR
END IF
RETURN Architecture
END
```

14. Applications of Mobile IP

- Mobile employees accessing organizational services while moving between networks.
- Continuous Internet access across different points of attachment.
- Emergency response and disaster recovery communication.
- Mobile healthcare and ambulance communication.
- Transportation and fleet monitoring.
- Field engineering and remote maintenance.
- Temporary communication networks.
- Location-independent access to network applications.

```
ALGORITHM SelectMobileIPApplication(Requirement)
INPUT:
Requirement – mobility and communication requirement
OUTPUT:
Suitable application category
BEGIN
IF Requirement requires continuous Internet access while moving THEN
RETURN MOBILE_INTERNET_ACCESS
ELSE IF Requirement involves emergency vehicle mobility THEN
RETURN EMERGENCY_COMMUNICATION
ELSE IF Requirement requires remote patient/device connectivity THEN
RETURN MOBILE_HEALTHCARE
ELSE IF Requirement involves moving vehicles or fleets THEN
RETURN TRANSPORTATION
ELSE
RETURN GENERAL_MOBILE_ACCESS
END IF
END
```

15. Routing Protocols in Mobile Networks

Routing protocols determine how packets travel through changing mobile topologies. For this unit, routing is studied using the two major approaches: proactive and reactive.

Routing State

```
STATE:
RouteTable[Destination] = {NextHop, HopCount, SequenceNumber, State}
State ∈ {VALID, INVALID, UNKNOWN}
NeighborTable = currently reachable neighbors
RouteRequestID = identifier for route discovery
RouteLifetime = validity period
```

16. Proactive Routing

A proactive protocol maintains route information before a packet transmission request occurs. This can reduce route-discovery delay but may create periodic control overhead.

```
ALGORITHM ProactiveRouteMaintenance(Node)
INPUT:
Node – mobile node
OUTPUT:
Updated routing table
BEGIN
INITIALISE RouteTable
PERIODICALLY DO
Broadcast current routing information
RECEIVE routing updates from neighbors
FOR each Update DO
Validate Update
FOR each Destination in Update DO
Calculate NewMetric
IF NewMetric < CurrentMetric(Destination) OR
Update.SequenceNumber > CurrentSequenceNumber(Destination) THEN
UPDATE RouteTable[Destination]
Set NextHop and HopCount
END IF
END FOR
END FOR
Remove expired or invalid entries
END PERIODICALLY
RETURN RouteTable
END
```

17. Proactive Routing – DSDV-Style Sequence-Number Update

A destination-sequenced distance-vector approach uses route metrics and sequence numbers to maintain fresh routes and reduce routing loops.

```
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)
INPUT:
Node – receiving node
ReceivedRoute = {Destination, NextHop, HopCount, SequenceNumber}
OUTPUT:
Updated routing table
BEGIN
Destination ← ReceivedRoute.Destination
NewHopCount ← ReceivedRoute.HopCount + 1
NewSequence ← ReceivedRoute.SequenceNumber
IF Destination not in RouteTable THEN
ADD ReceivedRoute with NextHop = Node.Neighbor
ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN
REPLACE RouteTable[Destination]
```

```
ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND
NewHopCount < RouteTable[Destination].HopCount THEN
REPLACE RouteTable[Destination]
END IF
RETURN RouteTable
END
```

18. Reactive Routing

A reactive protocol discovers a route only when a source needs to send data and no valid route is available. Route discovery and route maintenance are therefore demand-driven.

```
ALGORITHM ReactiveRouteDiscovery(Source, Destination)
INPUT:
Source – requesting node
Destination – required destination
OUTPUT:
Route or ROUTE_NOT_FOUND
BEGIN
IF RouteTable[Destination].State = VALID THEN
RETURN RouteTable[Destination]
END IF
RouteRequestID ← RouteRequestID + 1
Create RouteRequest(Source, Destination, RouteRequestID)
Broadcast RouteRequest to neighbors
WAIT for RouteReply or RouteFailure
IF RouteReply received THEN
Establish reverse path
Establish forward path
Mark RouteTable[Destination] as VALID
RETURN RouteTable[Destination]
ELSE
RETURN ROUTE_NOT_FOUND
END IF
END
```

19. Reactive Routing – AODV-Style Route Request

```
ALGORITHM ProcessRouteRequest(Node, Request)
INPUT:
Node – current receiving node
Request = {Source, Destination, RequestID, HopCount}
OUTPUT:
Route discovery action
BEGIN
IF Request already processed THEN
DISCARD Request
```

```
RETURN
END IF
Record reverse route toward Request.Source
Mark Request as processed
IF Node = Request.Destination THEN
Create RouteReply
SEND RouteReply toward Source
RETURN
END IF
IF ValidRouteTo(Request.Destination) THEN
Create RouteReply using valid route
SEND RouteReply toward Source
ELSE
Request.HopCount ← Request.HopCount + 1
FOR each Neighbor in NeighborTable DO
FORWARD Request TO Neighbor
END FOR
END IF
END
```

20. Reactive Route Maintenance

```
ALGORITHM MaintainReactiveRoute(Source, Destination)
INPUT:
Source and Destination
OUTPUT:
Route status
BEGIN
MONITOR active route
IF LinkFailureDetected THEN
Mark failed link as INVALID
IF AlternateValidRoute exists THEN
Select alternate route
RETURN ALTERNATE_ROUTE_USED
ELSE
Generate RouteError
SEND RouteError toward affected source
IF Source still requires communication THEN
Initiate new RouteDiscovery
ELSE
RETURN ROUTE_TERMINATED
END IF
END IF
END IF
RETURN ROUTE_ACTIVE
```

END

21. Proactive vs Reactive Routing

FEATURE	PROACTIVE	REACTIVE
Route availability	Maintained before data transmission	Obtained when required
Route discovery delay	Low	Present when route is unknown
Control overhead	Periodic/continuous	Demand-driven
Stale information	Possible	Route is generally discovered when needed
Frequent traffic	Often suitable	May repeatedly use discovered routes
Infrequent traffic	May waste control traffic	Often efficient
Topology changes	Routing tables updated periodically	Route repair/discovery triggered
Typical example	DSDV-style	AODV-style

22. Proactive or Reactive – Decision Algorithm

ALGORITHM SelectRoutingApproach(TrafficRate, MobilityRate, DelayRequirement, Bandwidth)
INPUT:
TrafficRate – frequency of data communication
MobilityRate – frequency of topology changes
DelayRequirement – acceptable route setup delay
Bandwidth – available bandwidth
OUTPUT:
Recommended routing approach
BEGIN
IF DelayRequirement = VERY_LOW AND TrafficRate = HIGH THEN
Recommendation ← PROACTIVE
ELSE IF TrafficRate = LOW AND Bandwidth = LIMITED THEN
Recommendation ← REACTIVE
ELSE IF MobilityRate = HIGH AND TrafficRate = INTERMITTENT THEN
Recommendation ← REACTIVE
ELSE IF TrafficRate = HIGH AND DelayRequirement = LOW THEN
Recommendation ← PROACTIVE
ELSE
Recommendation ← EVALUATE_BOTH
END IF
RETURN Recommendation
END

23. Route Selection and Metric Calculation

```
FUNCTION CalculateRouteMetric(HopCount, Delay, LossRate, EnergyCost)
INPUT:
HopCount, Delay, LossRate, EnergyCost
OUTPUT:
RouteMetric
BEGIN
RouteMetric ←
W1 × HopCount +
W2 × Delay +
W3 × LossRate +
W4 × EnergyCost
RETURN RouteMetric
END
ALGORITHM SelectBestRoute(RouteSet)
INPUT:
RouteSet – candidate routes
OUTPUT:
BestRoute
BEGIN
BestRoute ← NULL
BestMetric ← INFINITY
FOR each Route in RouteSet DO
Metric ← CalculateRouteMetric(Route.HopCount,
Route.Delay,
Route.LossRate,
Route.EnergyCost)
IF Metric < BestMetric THEN
BestMetric ← Metric
BestRoute ← Route
END IF
END FOR
RETURN BestRoute
END
```

24. Packet Forwarding in a Mobile Network

```
ALGORITHM ForwardMobilePacket(Node, Packet)
INPUT:
Node – current forwarding node
Packet – data packet
OUTPUT:
Delivery, forwarding or drop decision
BEGIN
Destination ← Packet.Destination
```

```
IF Node = Destination THEN
DELIVER Packet
RETURN DELIVERED
END IF
IF ValidRouteTo(Destination) THEN
NextHop ← RouteTable[Destination].NextHop
IF LinkStatus(NextHop) = UP THEN
FORWARD Packet TO NextHop
RETURN FORWARDED
ELSE
Mark route as INVALID
RETURN ROUTE_FAILURE
END IF
ELSE
RETURN ROUTE_DISCOVERY_REQUIRED
END IF
END
```

25. Routing Loop Detection

```
ALGORITHM DetectRoutingLoop(Path)
INPUT:
Path – ordered sequence of nodes
OUTPUT:
TRUE if loop exists, otherwise FALSE
BEGIN
Visited ← EMPTY_SET
FOR each Node in Path DO
IF Node IN Visited THEN
RETURN TRUE
END IF
ADD Node TO Visited
END FOR
RETURN FALSE
END
```

26. Manual Trace / Desk Check

Mobile routing pseudocode must be manually traceable. Record the node, current route state, received message, decision, next hop, and resulting route table state.

STEP	NODE	EVENT	DECISION	ROUTE STATE
1	S	Needs D	No valid route	UNKNOWN
2	S	Broadcast RREQ	Forward to neighbors	DISCOVERY
3	A	Receives RREQ	Record reverse route	DISCOVERY
4	D	Receives RREQ	Generate RREP	VALID

STEP	NODE	EVENT	DECISION	ROUTE STATE
5	S	Receives RREP	Install route S→A→D	VALID
6	S	Data packet	Forward to A	ACTIVE

27. What Will NOT Be Accepted as Mobile Routing Pseudocode?

Too vague:

Start
Find route
Send packet
Handle movement
Display result
Stop
There is insufficient computational logic.

Programming code disguised as pseudocode:

```
for(int i=0; i<n; i++) {
if(route[i].valid) send(packet);
}
```

Acceptable algorithmic form:

```
FOR each Route in RouteTable DO
IF Route.State = VALID THEN
FORWARD Packet TO Route.NextHop
RETURN FORWARDED
END IF
END FOR
RETURN ROUTE_DISCOVERY_REQUIRED
```

28. Programming-Language Syntax Should Not Be Used

- Avoid Java, Python, C, C++, Android API calls, or simulator-specific commands unless actual implementation code is explicitly requested.
- Use SEND, RECEIVE, FORWARD, DELIVER, DROP, UPDATE, ADD ROUTE and REMOVE ROUTE.
- Use ← for assignment.
- Use explicit network conditions instead of vague statements such as 'handle mobility'.

29. Comments

```
// Detect whether the current link is still usable
IF LinkStatus(NextHop) = DOWN THEN
Mark Route INVALID
Initiate route repair
END IF
```

Comments may clarify a non-obvious networking decision, but comments cannot replace the routing logic. The supplied DAA standard similarly requires the pseudocode itself to analyse how the result is obtained.

30. Quality Checklist

- Inputs are clearly defined.
- Outputs are clearly defined.
- Mobile-network states have clear meanings.
- Addresses, routes and sequence information are initialised.
- Packet forwarding conditions are explicit.
- Registration and handoff conditions are explicit.
- Route discovery and route repair have clear termination.
- Link failure and unavailable-route cases are addressed.
- The algorithm can be manually traced.
- Programming-language-specific syntax is avoided.
- Another programmer could implement the algorithm without guessing the logic.

31. How Mobile Computing Pseudocode Will Be Assessed

CRITERION	WHAT IS EXPECTED
Algorithmic Correctness	The mobility or routing logic correctly solves the stated problem.
Completeness	Inputs, outputs, state, decisions, loops, packet actions and termination are sufficiently specified.
Precision	Conditions such as route validity, link state, registration status and sequence number are explicit.
Networking Understanding	The solution correctly reflects Mobile IP, cellular/ad hoc networking or routing concepts.
Appropriate Abstraction	Network operations are represented clearly without unnecessary programming syntax.
Traceability	A small network scenario can be manually executed using the pseudocode.
Technical Justification	The selected proactive/reactive or mobility mechanism is supported by network characteristics.

32. Final Test

Before submitting any Mobile Computing pseudocode, ask:

- Can another competent programmer implement my algorithm without asking what I meant?
- Can the same algorithm be implemented in Java, Python, C/C++ or a network simulator without changing its underlying logic?
- Does the algorithm explicitly handle mobility, routing state and failure conditions?

- Can I trace the algorithm using a small mobile-network example?

33. THE MOBILE COMPUTING DESIGN RULE

UNDERSTAND → MODEL NETWORK → DEFINE STATE → DESIGN ALGORITHM → WRITE PSEUDOCODE → TRACE → TEST → REFINE → IMPLEMENT

Do not write the program first and convert it into pseudocode afterwards. Design the mobile-network behaviour first. The program or simulator is the implementation of that design. This follows the source handbook's final design principle: THINK → DESIGN → WRITE PSEUDOCODE → TRACE → REFINE → IMPLEMENT.

ANSWERS

ANSWER TO Q1 – Wireless Network Architecture and Mobility Analysis (6 Marks)

Part A – Introduction to Wireless Networks

Shared Group Work

Characteristics and components of wireless networks

A wireless network connects devices without cables, using radio waves as the transmission medium. Its main characteristics are: (1) a shared broadcast medium, so that all stations within range compete for the same channel and interfere with each other; (2) limited and variable bandwidth (for example, an IEEE 802.11 access point with a nominal rate of 54 Mb/s gives a real throughput of only about 25 Mb/s, which is shared between all users); (3) higher error rates than cables (bit error rates of 10^{-3} to 10^{-6} against 10^{-9} or better in wired links) because of fading, noise and interference; (4) mobility of the users, so the point of attachment changes; (5) limited battery energy and processing power of the devices; and (6) weaker security, because the signal can be received by anyone in range.

Main components: Mobile Nodes (MN) such as phones, laptops, tablets and vehicle units; wireless access points / base stations (AP/BS) that provide radio coverage and connect the wireless side to the wired network; the wireless communication links (radio channels) between MN and AP; a wired backbone (switches, routers and gateways); and servers or correspondent hosts that provide the services.

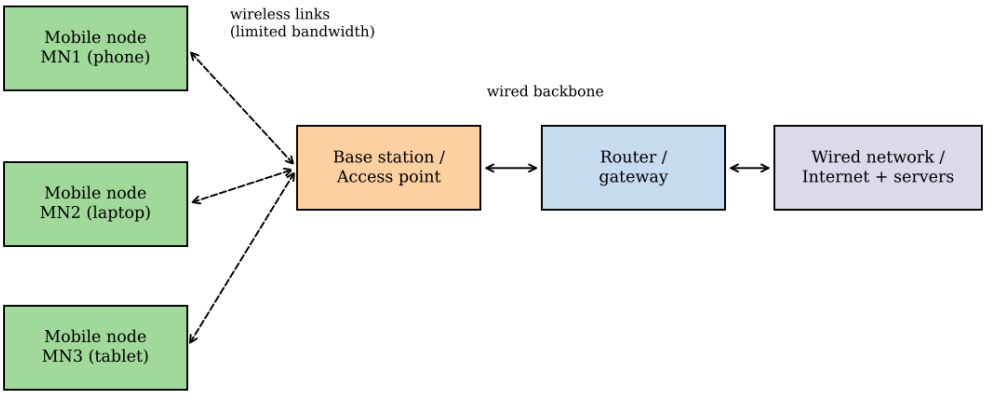


Figure 1: Infrastructure-based wireless network architecture

Bandwidth, connectivity and mobility constraints

- Bandwidth: the capacity of a cell is shared. If an AP delivers 25 Mb/s of real throughput and 10 users are active, each user gets about 2.5 Mb/s on average; with 50 users it is only 0.5 Mb/s.
- Connectivity: the signal strength falls with distance. The free-space path loss is FSPL (dB) = 20 log10(d) + 20 log10(f) + 32.44 (d in km, f in MHz). For f = 2,400 MHz the path loss is 74.0 dB at 50 m, 80.0 dB at 100 m, 86.1 dB at 200 m and 94.0 dB at 500 m

(about 6 dB more for each doubling of the distance); walls and obstacles add more, so connectivity is limited to the coverage area of the AP.

- Mobility: when the user leaves the coverage area, the node must attach to another AP/BS (handoff), and a short interruption of the communication occurs during that handoff.

Communication between mobile users and the infrastructure (simulated scenario)

Scenario: a mobile node MN1 sends a voice stream (G.711, 160-byte payload every 20 ms = 50 packets per second) to a server through an access point. The communication flow tested was: (1) MN1 scans the channels and finds AP1 (beacon); (2) authentication and association with AP1 (about 5 ms); (3) the node obtains an IP address by DHCP (about 20 ms); (4) packets MN1 → AP1 → router → server, with replies in the opposite direction; (5) when the signal falls, MN1 scans for AP2 and re-associates. The resulting timing of one packet is: wireless transmission (200 bytes at 2 Mb/s effective = 0.8 ms plus MAC access and retransmissions about 7 ms), AP to router 1 ms, wired backbone 20 ms and server 5 ms, i.e. a one-way delay of about 33 ms, well below the 150 ms limit for voice. The test confirms the flow MN → AP → wired network → server and back.

Student A – Network Architecture Analysis

Major network components and their roles

Component	Role
Mobile node (MN)	End device that generates and receives data and moves between coverage areas; has a radio interface, limited battery and an IP address
Access point / base station	Provides radio coverage, controls access to the channel, authenticates users, and bridges the wireless link to the wired network; each one covers a cell
Wireless link	Radio channel between MN and AP; shared, error-prone, limited bandwidth
Router / gateway	Forwards packets between the access network and the Internet; hosts firewall and NAT
Wired backbone and servers	Carry traffic at high speed; servers and correspondent nodes provide the application

Wireless communication characteristics and their effect

Limited bandwidth: the available rate falls as the distance grows (rate adaptation: 54 Mb/s near the AP, 6 Mb/s at the cell edge) and as the number of users grows, which raises the delay because packets wait for the channel. Wireless connectivity: when the received signal falls below the receiver sensitivity (for example −85 dBm) the link is lost; in the cell edge region the error rate and retransmissions grow, increasing the delay and the loss. The wireless hop is therefore usually the bottleneck of the whole path, and protocols and applications should be tolerant to losses that are not caused by congestion.

How mobile nodes communicate through the infrastructure

A mobile node never communicates directly with another distant node: every packet is sent first to the access point (or base station) of its cell, which relays it to the wired network; the infrastructure forwards it to the destination AP, which transmits it to the destination MN over the wireless link. The AP also performs medium access control (CSMA/CA in Wi-Fi, scheduling in cellular networks), so the AP is the central point of the communication, and all mobility functions (association, handoff, authentication) are executed with the help of the infrastructure.

Student B – Mobility and Performance Analysis

Representative mobility scenarios, handoff and interruption

Four mobility scenarios were defined for cells whose access points are 1,000 m apart: stationary user, pedestrian (5 km/h), urban vehicle (50 km/h) and highway vehicle (100 km/h). A handoff occurs each time the user crosses into the next cell: the time between handoffs is 1,000 m divided by the speed. The interruption of the connection during a handoff is the sum of the scanning, authentication/association and re-routing times; values of 50 ms (fast handoff), 200 ms (typical) and 500 ms (slow handoff) were used. For the voice stream of 50 packets per second, the number of packets lost in each handoff is 50 × interruption: 2.5, 10 and 25 packets.

Speed	Handoffs per minute	Time between handoffs	Loss at 50 ms	Loss at 200 ms	Loss at 500 ms
5 km/h	0.08	720 s	0.007%	0.028%	0.069%
20 km/h	0.33	180 s	0.028%	0.111%	0.278%
50 km/h	0.83	72 s	0.069%	0.278%	0.694%
100 km/h	1.67	36 s	0.139%	0.556%	1.389%

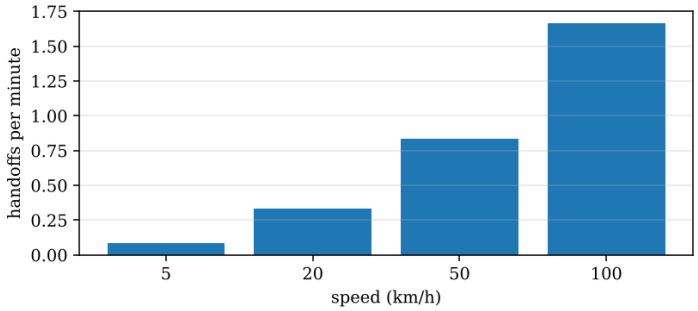


Figure 2: Number of handoffs per minute as a function of speed (cell spacing 1,000 m)

Packet loss and transmission delay; effect of changing network conditions

The packet loss caused by the handoffs alone is small (below 1.4%) but it comes in bursts of 2.5 to 25 consecutive packets, which is audible in a voice call and which TCP interprets as congestion. Transmission delay at the wireless hop is L/R: a 1,000-byte packet needs 1000 × 8 / 54,000,000 = 0.15 ms at 54 Mb/s but 1000 × 8 / 6,000,000 = 1.33 ms at 6 Mb/s, and the contention delay grows with the number of users: if the one-way wireless delay is about 7 ms for 5 active users it can be 30 ms or more with 40 users. If the signal quality falls (cell edge,

obstacles) the rate drops, retransmissions occur and both the delay and the loss increase; if the number of users increases, the available bandwidth per user falls.

Comparison of communication behaviour under different mobility conditions

Condition	Handoff frequency	Connectivity interruption	Packet loss	Delay behaviour
Stationary	None	None	Only radio errors	Stable
Pedestrian (5 km/h)	Very rare (every 12 min)	Rare short gaps	Very low	Stable, small variation
Vehicle (50 km/h)	Every 72 s	Repeated gaps of 0.05–0.5 s	0.07–0.7%, bursty	Peaks at each handoff
Highway (100 km/h)	Every 36 s	Frequent; overlap only 3.6 s	0.14–1.4%, bursty	Frequent peaks; risk of failed handoff

Part B – Wireless Mobility Scenario

Shared Group Work

Scenario: a mobile user in a vehicle at 50 km/h (13.9 m/s) moves along a road through two wireless cells (Figure 3). BS1 and BS2 are 1,000 m apart, each cell has a radius of 500 m, and the coverage areas overlap by 100 m (the handoff zone P3). A voice session of 50 packets per second is active during the movement.

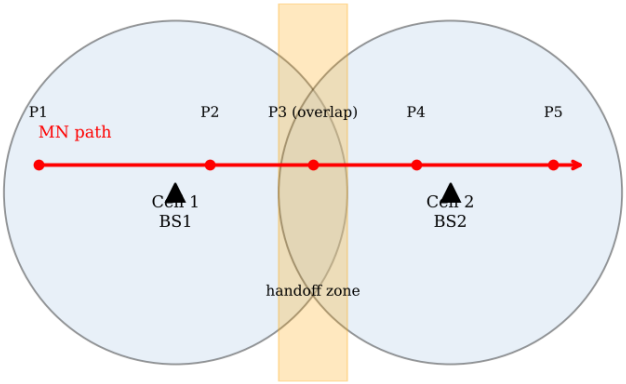


Figure 3: Mobility scenario – a mobile user moving between two coverage areas (positions P1 to P5)

Student A – Analytical Investigation

Successful and interrupted communication

- Successful communication: the packets are delivered to the destination with a delay below 150 ms, a loss below 1% and without a break longer than about 100 ms. This is the case at P1, P2, P4 and P5 (inside a cell with a good signal) and also at P3 if the handoff is completed quickly.
- Interrupted communication: the mobile node has no valid attachment, e.g. between detaching from BS1 and completing the association with BS2 (about 200 ms), or if the

signal of the old cell drops before the new cell is available (the handoff fails and the connection is lost).

Sequence of mobility events and delay/packet-delivery behaviour

Event	Position / time	Signal and action	Communication effect
1	P1, t = 0 s	Associated with BS1, RSSI −60 dBm	Normal: delay about 33 ms, no loss
2	P2, t ≈ 30 s	RSSI of BS1 falls to −75 dBm: scanning starts	Normal; slight rate reduction
3	P3, enters overlap (t ≈ 36 s)	RSSI(BS2) exceeds RSSI(BS1) by the 5 dB hysteresis: handoff decision	Handoff starts
4	P3, handoff (200 ms)	Authentication and association with BS2, new route to the MN	Interruption: 10 packets lost (50 pps × 0.2 s)
5	P3 + 0.2 s	MN attached to BS2, transmission resumes	Delay peak about 60 ms, then normal
6	P4 to P5	Normal operation in cell 2	Normal: delay about 33 ms

Increasing mobility conditions

The time the user spends in the overlap zone is t = 100 m / speed. The handoff needs time for scanning, decision and association; with a total requirement of 1.5 s the handoff is safe only if the overlap time is at least about twice this value.

Speed	Overlap time (100 m)	Ratio to the 1.5 s requirement	Packets lost per handoff (200 ms)	Result
5 km/h	72.0 s	48	10	Safe
50 km/h	7.2 s	4.8	10	Safe
100 km/h	3.6 s	2.4	10	Safe, small margin
200 km/h	1.8 s	1.2	10	Risky, may fail
300 km/h	1.2 s	0.8	connection lost	Handoff fails: session interrupted

Analytical evidence: at 50 km/h the user makes 0.83 handoffs per minute, loses 10 packets per handoff (0.28% of the packets), and the useful transmission time is 1 − (0.83 × 0.2 s)/60 s = 99.72%. At 100 km/h the loss doubles to 0.56%, the margin of the overlap zone falls to 2.4, and beyond about 240 km/h (overlap time equal to the 1.5 s requirement) handoffs begin to fail unless the overlap is increased or the handoff is made faster.

Student B – Mobility Logic

Mobility decision sequence and the change of the point of attachment

The mobile node measures the signal strength of the serving station and of the neighbouring stations. It changes its point of attachment only when three conditions are satisfied together: (1) the serving signal is below a threshold (−75 dBm), (2) a candidate station is stronger than the serving station by a hysteresis margin of 5 dB, and (3) the candidate has stayed better for a minimum dwell time (for example 1 s). The hysteresis and the dwell time prevent the ping-pong effect, in which a node at the cell boundary would switch repeatedly between the two stations and waste time in repeated handoffs.

ALGORITHM HandoffDecision(MN, ServingBS, CandidateSet, THRESHOLD, HYSTERESIS, MIN_DWELL)
INPUT:
MN – mobile node; ServingBS – current point of attachment
CandidateSet – neighbouring stations with measured signal strength RSSI
THRESHOLD = −75 dBm; HYSTERESIS = 5 dB; MIN_DWELL = 1 second
OUTPUT:
Decision ∈ {STAY, HANDOFF_TO(station), CONNECTION_LOST}
BEGIN
BestBS ← NULL; BestRSSI ← −INFINITY
IF RSSI(ServingBS) < MIN_USABLE_SIGNAL AND CandidateSet is empty THEN
START TIMER lostTimer
IF lostTimer EXPIRES THEN RETURN CONNECTION_LOST
END IF
IF RSSI(ServingBS) ≥ THRESHOLD THEN
RETURN STAY
END IF
FOR each Candidate in CandidateSet DO
IF RSSI(Candidate) > RSSI(ServingBS) + HYSTERESIS AND RSSI(Candidate) > BestRSSI THEN
BestBS ← Candidate; BestRSSI ← RSSI(Candidate)
END IF
END FOR
IF BestBS = NULL THEN
RETURN STAY
END IF
IF BestBS has satisfied the conditions for at least MIN_DWELL THEN
RETURN HANDOFF_TO(BestBS)
ELSE
RETURN STAY
END IF
END

Communication conditions affected, termination condition and efficiency

During a handoff the following are affected: the link layer attachment (the old association is released), the IP routing to the node (the path of the packets changes), the delay and the packet loss (10 packets are lost per handoff at 200 ms), the throughput (rate adaptation in the new cell) and, for TCP sessions, the congestion window (the loss is interpreted as congestion). The scenario is completed when the mobile node reaches the final position P5 with an active attachment and all the packets of the session have been delivered or acknowledged; it is terminated abnormally if the node has no valid attachment for longer than the timeout (5 s) or if the handoff fails (CONNECTION_LOST). Relation between repeated mobility and efficiency: efficiency = 1 − (handoffs per minute × interruption time)/60. For 0.83 handoffs per minute and 0.2 s this is 99.72%, but with ping-pong (for example 6 handoffs per minute) it would fall to 98%, so frequent attachment changes reduce the efficiency in proportion to the handoff rate; therefore hysteresis, larger overlap and fast handoff are important.

Individual Verification (Q1)

Student A should be able to explain Figure 1 and the table of the components, and Student B the scenario of Figure 3, the handoff decision and the numbers above; each can answer the other's part because the same formulas are used: loss per handoff = packet rate × interruption time, time between handoffs = cell spacing / speed and overlap time = overlap length / speed.

ANSWER TO Q2 – Mobile IP and Mobility Management (7 Marks)

Shared Group Work

Mobile IP architecture and terminology

Mobile IP (RFC 5944 for IPv4) lets a mobile node keep its permanent home address while it moves between networks, so that TCP connections and applications continue without change. Terminology: Home address (HoA) – the permanent IP address of the MN in its home network; Home network – the network of the HoA; Home Agent (HA) – a router in the home network that keeps the binding between the HoA and the current care-of address and tunnels packets to the MN; Foreign network – the visited network; Foreign Agent (FA) – a router in the foreign network that offers a care-of address and forwards packets to the visiting MN; Care-of address (CoA) – the address that identifies the current location of the MN (the FA address, or a co-located address obtained by the MN); Correspondent node (CN) – the host that communicates with the MN; Binding – the association HoA ↔ CoA with a lifetime; Tunnel – the encapsulation of the original packet in a new IP packet addressed to the CoA.

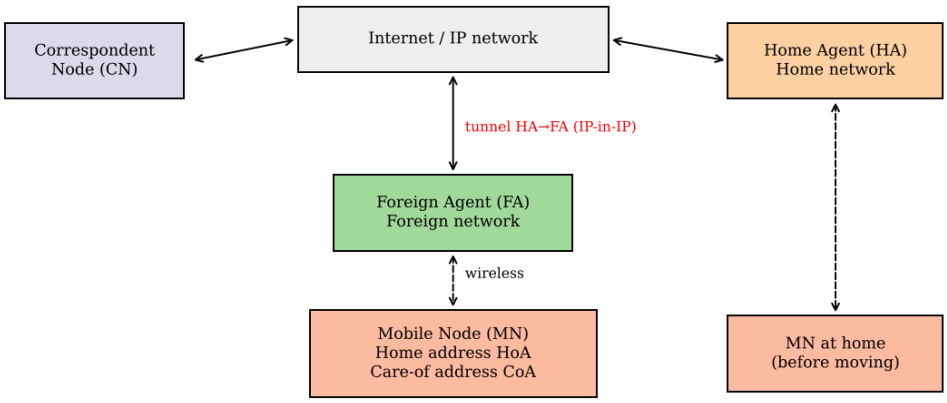


Figure 4: Mobile IP architecture (MN, HA, FA, CN and the tunnel)

Agent discovery, registration and packet delivery

Agent discovery: the HA and FA periodically broadcast agent advertisements (ICMP router advertisements with a mobility extension, usually once per second, lifetime 3 s); the MN can also send an agent solicitation. By comparing the network prefix of the advertisement with its home prefix the MN finds out whether it is at home or visiting a foreign network; if it misses three advertisements it concludes that it has moved. Registration: in a foreign network the MN obtains the CoA (the FA address advertised) and sends a Registration Request (HoA, HA address, CoA, lifetime, identification field, authentication extension) to the FA, which relays it to the HA; the HA checks the authentication, creates or updates the binding and returns a Registration Reply with a result code and the granted lifetime, which the FA passes to the MN. Packet delivery: the CN sends packets to the HoA; they are routed to the home network, where the HA intercepts them (proxy ARP), encapsulates them (IP-in-IP) with the CoA as the outer destination and tunnels them to the FA, which decapsulates and delivers them to the MN. The packets from the MN to the CN can go directly (with the HoA as the source) or via the reverse tunnel.

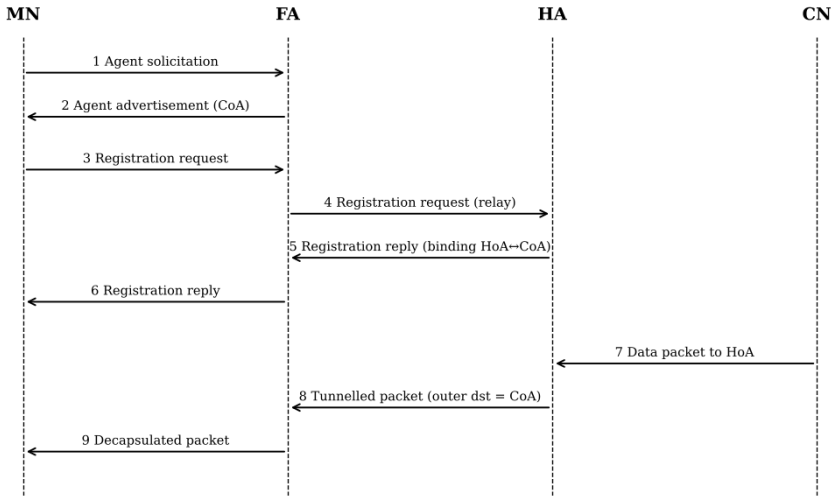


Figure 5: Registration and packet-flow sequence in Mobile IP

Worked example of Mobile IP communication

Addresses: home network 10.1.1.0/24, MN home address HoA = 10.1.1.20, HA = 10.1.1.1; foreign network 192.168.5.0/24 with FA = 192.168.5.1 (CoA = 192.168.5.1); CN = 172.16.4.9. One-way delays: MN–FA 5 ms, FA–HA 40 ms, CN–HA 30 ms, CN–FA 25 ms. A voice stream of 50 packets per second (160-byte payload, 40 bytes of RTP/UDP/IP header) runs from the CN to the MN.

Step	Action	Addresses	Time (ms)
1	MN moves to the foreign network and receives the agent advertisement of the FA (advertisement every 1 s: average wait 500 ms)	FA 192.168.5.1	500
2	MN sends Registration Request to the FA, the FA relays it to the HA	MN→FA 5 ms, FA→HA 40 ms	45
3	HA authenticates, creates the binding 10.1.1.20 ↔ 192.168.5.1 and replies	HA → FA → MN	45
4	CN sends packet: source 172.16.4.9, destination 10.1.1.20	CN → home network	—
5	HA intercepts the packet and encapsulates it: outer source 10.1.1.1, outer destination 192.168.5.1	IP-in-IP, 20 bytes added	tunnel 40
6	FA decapsulates and delivers the original packet to the MN	FA → MN 5 ms	5

Results: the interruption of the communication caused by the move is 500 + 45 + 45 = 590 ms (about 0.59 s); at 50 packets per second this means about 29.5 lost packets (about 30). The delay from the CN to the MN through the tunnel is 30 + 40 + 5 = 75 ms, compared with 25 + 5 = 30 ms for a direct path.

Student A – Mobile IP Operation

Roles of the four entities

- Mobile Node: keeps its permanent HoA, detects that it has moved from the agent advertisements, obtains the CoA, registers with the HA through the FA, re-registers before the lifetime expires and sends/receives data using its HoA so that the CN is not aware of the move.
- Home Agent: stays in the home network, maintains the binding table (HoA → CoA, lifetime), intercepts the packets addressed to the HoA, encapsulates and tunnels them to the CoA, and authenticates the registration requests.
- Foreign Agent: advertises itself and the CoA, relays the registration messages, keeps a visitor list, decapsulates the tunnelled packets and delivers them to the visiting MN (and may act as the default router of the MN).
- Correspondent Node: any host that communicates with the MN; in basic Mobile IP it sends to the HoA and does not know the mobility; with route optimisation it can keep its own binding cache and send directly to the CoA.

Registration, location updating, packet delivery and tunnelling

ALGORITHM ProcessRegistrationRequest(HA, Request)
INPUT:
HA – home agent; Request = {HomeAddress, CareOfAddress, Lifetime, Identifier, Authenticator}
OUTPUT:
Registration reply code (ACCEPTED or the reason of rejection)
BEGIN
IF Request.HomeAddress is not served by HA THEN
SEND RegistrationReply(UNKNOWN_HOME_ADDRESS) TO Request.Source
RETURN REJECTED
END IF
IF Authenticator is not valid for MN.SecurityAssociation THEN
SEND RegistrationReply(AUTHENTICATION_FAILED)
RETURN REJECTED
END IF
IF Request.Identifier ≤ LastIdentifier[Request.HomeAddress] THEN
SEND RegistrationReply(REPLAY_DETECTED)
RETURN REJECTED
END IF
GrantedLifetime ← MIN(Request.Lifetime, MAX_LIFETIME)
IF Request.CareOfAddress = Request.HomeAddress OR GrantedLifetime = 0 THEN
REMOVE ROUTE Binding[Request.HomeAddress]
ELSE
UPDATE ROUTE Binding[Request.HomeAddress] ← {Request.CareOfAddress, CurrentTime + GrantedLifetime}
START TIMER bindingTimer FOR GrantedLifetime

END IF
LastIdentifier[Request.HomeAddress] ← Request.Identifier
SEND RegistrationReply(ACCEPTED, GrantedLifetime)
RETURN ACCEPTED
END

Location updating is the registration itself: each time the MN changes its CoA, it sends a new Registration Request, and the HA replaces the old binding. When the lifetime (for example 600 s) is about to expire, the MN re-registers. Packet delivery and tunnelling follow the algorithm TunnelPacketToMobileNode of the standard (Section 9): the HA looks up Binding[HoA], encapsulates the packet with the CoA as the outer destination and forwards it; if no binding exists, the packet is dropped or delivered in the home network.

Student B – Mobile IP Challenge Analysis

Challenge	Analysis (values of the worked example)	Reduction
Registration delay	Registration round trip MN–FA–HA = 2 × (5 + 40) = 90 ms; plus movement detection (average 500 ms with 1 s advertisements) gives 590 ms of interruption and about 30 lost packets at 50 pps	Advertisement interval of 0.2 s (detection 100 ms; total about 190 ms, about 10 packets); link-layer triggers; HA closer to the user (regional registration)
Tunnelling overhead	IP-in-IP adds 20 bytes to every packet: 20/200 = 10% for a 160-byte voice packet (with 40 bytes of headers), but only 20/1,440 = 1.4% for a 1,400-byte data packet; minimal encapsulation adds only 8 to 12 bytes	Header compression, minimal encapsulation, route optimisation for small packets
Triangular routing	CN → HA → FA → MN takes 30 + 40 + 5 = 75 ms against 25 + 5 = 30 ms directly: 2.5 times longer, and the HA carries all the traffic	Route optimisation (binding cache at the CN, as in Mobile IPv6), reverse tunnelling is not needed if there is no ingress filtering
Packet loss during mobility	Packets sent to the old CoA during the 0.59 s of handoff are lost (about 30 at 50 pps); TCP sees the loss as congestion	Smooth handoff: the old FA buffers and forwards packets to the new FA (binding update), bicasting
Security	The registration can be forged to redirect the traffic of the MN (hijacking), replayed, or the tunnel can be eavesdropped	Authentication extensions MN–HA (keyed MD5 / HMAC with SPI), identification field (timestamp or nonce) against replay, IPsec for the tunnel

Challenge	Analysis (values of the worked example)	Reduction
Scalability	One HA handles many MNs: 10,000 mobile nodes that move every 60 s generate 10,000/60 ≈ 167 registrations per second, and the HA processes all the tunnelled traffic; it is a bottleneck and a single point of failure	Several HAs (load sharing), hierarchical or regional registration, longer lifetimes, local mobility anchors

Joint Design Question

Question: if a mobile user frequently changes its network of attachment during an active communication session, what problems can occur in Mobile IP and how can they be reduced?

Problems: (1) each change needs movement detection plus registration (590 ms in the example), so for a user that changes network every 20 s the session is interrupted 590 ms/20 s = 2.95% of the time and about 30 packets are lost every time, with TCP slow start after each loss; (2) the signalling load grows with the number of moves: every move produces a request and a reply that travel to the HA, which can be far away (80 to 90 ms round trip), and the HA must process the registrations of many users; (3) triangular routing keeps the traffic going through the home network, adding delay (75 ms instead of 30 ms) and load on the HA; (4) packets that are already in the tunnel to the old CoA are lost; (5) frequent re-registration increases the exposure to attacks and the energy use of the battery-powered node; (6) if the registration fails or the reply is lost, the node is disconnected until the next attempt.

Problem	Proposed reduction	Expected result
Long detection and registration delay	Shorter advertisement interval (0.2 s), layer-2 triggers (link-up events), fast handoff with pre-registration through the old FA	Interruption from 590 ms to about 100–190 ms; loss from 30 to 5–10 packets
Registration signalling to a far HA	Hierarchical / regional registration: the MN registers locally with a gateway FA while moving inside a region, and the HA is updated only when the region changes	Local registrations take about 10–20 ms; HA load is reduced by the factor of the moves inside regions
Triangular routing	Route optimisation: the CN keeps a binding and sends directly to the CoA	Delay 30 ms instead of 75 ms; HA relieved
Loss of packets in transit	Buffering and forwarding by the old FA; bicasting during handoff	Almost no loss for the packets in transit
Security overhead and attacks	Lifetime-based binding, authentication with HMAC, replay protection	Safe registrations at a small cost per move

Justification: the proposed improvement combines a fast handoff (smaller detection and registration time), regional registration (fewer and shorter signalling exchanges) and route optimisation (shorter paths). With 0.2 s advertisements the interruption per move falls from 590 to about 190 ms, i.e. the fraction of time lost for a user that moves every 20 s falls from

2.95% to 0.95%, and packets lost per move from 30 to about 10; with buffering at the old FA the loss in transit practically disappears.

Cross-Understanding (Q2)

Student A can explain the challenges through the table above (registration delay 90 ms + detection, tunnelling overhead 20 bytes, triangular routing 75 ms against 30 ms, packet loss, security and scalability). Student B can explain the complete process through Figure 5: advertisement → registration request via FA → HA authentication and binding → reply; and then CN → HoA → HA → tunnel to CoA → FA → MN.

ANSWER TO Q3 – Cellular Networks versus Ad Hoc Wireless Networks (7 Marks)

Shared Group Work

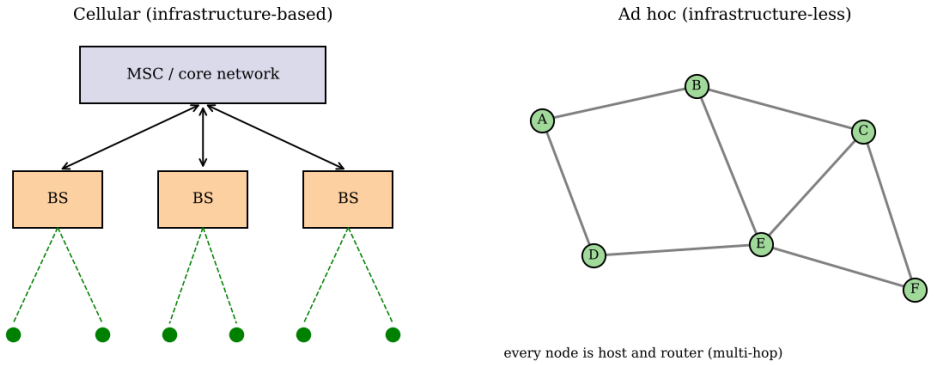


Figure 6: Cellular (infrastructure-based) and ad hoc (infrastructure-less) wireless networks

Cellular architecture: the service area is divided into cells, each served by a base station (BS); the base stations are connected to a mobile switching centre (MSC) / core network, which connects the network to the public telephone and data networks and holds the location databases (HLR/VLR) for authentication and mobility management. All communication passes through the BS and the core network (infrastructure-based). Ad hoc architecture: a set of mobile nodes communicate directly by radio without any fixed infrastructure; every node acts both as a host and as a router, and when two nodes are not within radio range, packets are forwarded by intermediate nodes (multi-hop). The topology is dynamic because the nodes move, switch off or fail.

Comparative analysis

Parameter	Cellular Network	Ad Hoc Network
Infrastructure	Fixed base stations, MSC, backbone and spectrum planning are required	None; the nodes themselves form the network
Communication	Centralised: through the BS and core network; the BS controls access to the channel	Distributed: direct between neighbours and multi-hop through intermediate nodes; contention-based medium access
Routing	Infrastructure-assisted: the network knows the location of the user (HLR/VLR) and routes the call	Distributed multi-hop routing protocols (proactive such as DSDV, reactive such as AODV)
Mobility	Managed by the network: location registration, paging, handoff; users move inside a planned structure	Nodes move freely; mobility changes the links and the routes continuously
Topology	Planned, static cell structure (frequency reuse, for example cluster size 7)	Dynamic, unpredictable, changes every few seconds

Parameter	Cellular Network	Ad Hoc Network
Scalability	Scales well by cell splitting, sectoring and more base stations; the cost is the infrastructure	Per-node throughput falls as the number of nodes and the hop count grow; routing overhead grows
Handoff	Hard or soft handoff controlled by the BS/MSC between cells	No handoff in the cellular sense; routes are re-established when links break
Failure handling	Failure of a BS or MSC removes the service of an area; protected by redundancy and overlapping cells	Failure of a node is tolerated by finding another route; a partition occurs if no path remains
Applications	Public voice and mobile broadband, mobile Internet, IoT, banking	Disaster recovery, military, vehicular networks, temporary and field networks, sensor networks, conferences

Student A – Cellular Network Investigation

- Role of the base stations and infrastructure: the BS provides the radio coverage of its cell, allocates channels, controls power and access, authenticates the user with the help of the core network and connects the radio side to the wired backhaul; the MSC switches calls, keeps the location databases and controls the handoff; the backbone gives the connection to the other networks.
- Centralised communication: even two phones in the same cell communicate through the BS; control, authentication, billing and resource allocation are centralised, which gives predictable quality and security at the price of dependence on the infrastructure.
- Handoff between cells: when the signal of the serving BS falls and a neighbouring BS is stronger, the MSC transfers the call. Hard handoff: break-before-make (a short interruption); soft handoff (CDMA): make-before-break (the connection is held by two BS during the change).
- Scalability and coverage: frequency reuse allows the same channels in distant cells: for the cluster size $N = i^2 + ij + j^2$ (for $i = 2$, $j = 1$ we get $N = 7$) the reuse distance is $D = R\sqrt{(3N)} = 4.58\ R$, and a system with 420 channels gives $420/7 = 60$ channels per cell; capacity is increased by cell splitting (smaller R), sectoring and more base stations.
- Suitable applications: public voice and video calls, mobile broadband Internet, mobile banking and payment, mobile healthcare when coverage exists, IoT and vehicle connectivity in covered areas.

Student B – Ad Hoc Network Investigation

- Architecture: an arbitrary graph of mobile nodes (Figure 6, right) with links between nodes inside the radio range; no BS, no central control; each node is host and router.
- Communication between mobile nodes: neighbours communicate directly; if the destination is out of range, the source obtains a route (by a routing protocol) and the packets are relayed by intermediate nodes. For example, in Figure 6 node A reaches F by A → B → E → F (3 hops).

- Dynamic topology: nodes move, switch on and off and the radio links appear and disappear, so the routes change. In the simulation of Q7 (30 nodes, 10 m/s) a node loses and gains neighbours several times per minute, and the control traffic of the routing protocol is required to follow these changes.
- Multi-hop communication: it extends the coverage beyond one radio range and saves transmission power, but every additional hop adds delay (about 10 ms per hop in the simulation), consumes channel capacity (the same packet is transmitted several times) and increases the chance of loss.
- Absence of fixed infrastructure: the network can be deployed in minutes and has no single point of failure, but there is no central authentication, address management or resource control; security, addressing and energy have to be solved in a distributed way.
- Suitable applications: emergency and disaster communication, military and battlefield networks, vehicular ad hoc networks (VANET), sensor networks, temporary meetings and events, and rural areas without infrastructure.

Joint Engineering Decision

Question: for an emergency/disaster environment where fixed infrastructure may be unavailable, which network architecture would you select and why? Decision: a mobile ad hoc network (MANET), preferably with one or more gateway nodes (satellite or cellular uplink) that can connect the ad hoc network to the Internet when a link is available. Reasons: (1) it does not depend on base stations or power and backhaul that a disaster may destroy; (2) it can be deployed immediately by the rescue teams (DeploymentTime = IMMEDIATE, InfrastructureAvailability = UNAVAILABLE in the standard algorithm of Section 13, which returns AD_HOC); (3) multi-hop relaying extends the coverage over the damaged area; (4) the failure of one node does not stop the network because another route can be used; (5) the cost and the set-up time are low. Cellular service is preferred only when the infrastructure is available and the coverage requirement is large.

Cross-Defence (Q3)

Student A – why cellular networks depend on infrastructure: the base station gives the only access to the radio channel (a phone cannot talk to another phone directly), allocates the frequencies and controls the power so that interference is avoided; the core network authenticates the user, knows where the user is (HLR/VLR), switches the call and performs the handoff; the backhaul connects the cells with each other and with the Internet. If the BS, the MSC or the backhaul or the electric power fails, the users of that area have no service although their phones work.

Student B – how ad hoc networks continue communication without fixed infrastructure: each node keeps information about its neighbours and a routing protocol builds routes through other nodes. If, in Figure 6, the link A–B breaks, A uses another neighbour: the path A → D → E → F is used instead of A → B → E → F; the routing protocol detects the failure (hello messages or link-layer feedback), reports the broken link, and finds a new route in a fraction of a second to a few seconds. Communication continues as long as at least one path exists in the connectivity graph of the moving nodes.

ANSWER TO Q4 – Applications of Mobile IP and Routing Requirements (7 Marks)

Shared Group Work

Real-world applications that need mobility support: continuous Internet access for mobile employees, emergency and disaster communication, mobile healthcare and ambulance communication, intelligent transportation and fleet monitoring, field engineering and remote maintenance, and temporary networks. Selected application: mobile healthcare with an ambulance that transmits the patient's vital signs and receives instructions from the hospital while it moves between networks.

Mobile entities: the ambulance communication unit (MN) with the patient monitor, the paramedic's tablet and the voice/video unit. Communication endpoints (correspondent nodes): the hospital electronic health record server, the emergency physician and the specialist at the hospital. Communication requirements: continuous IP connectivity with the same address while moving across cellular and Wi-Fi networks, low delay for voice and video, reliable delivery of the vital signs, security and privacy of the patient data.

Student B – Application Architecture

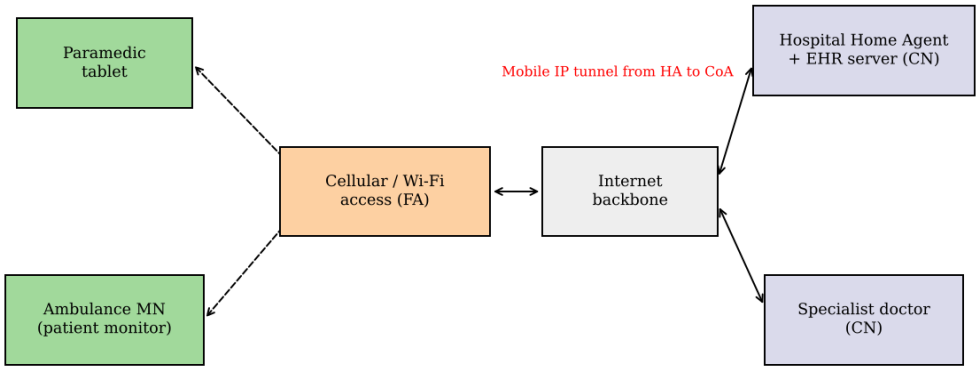


Figure 7: Application architecture – Mobile IP for ambulance communication

- Description: the hospital network is the home network of the ambulance unit; the hospital router acts as the Home Agent. In each visited area (cellular network gateway or Wi-Fi hotspot) a foreign agent (or a co-located CoA from DHCP) provides the care-of address.
- Mobile nodes and endpoints: MN = ambulance unit and tablet (home addresses in the hospital network); CN = EHR server and specialist workstation.
- How Mobile IP supports mobility: the ambulance keeps its home address, so the hospital applications always send to the same address; the HA tunnels the packets to the current CoA after each registration. The TCP sessions of the patient data and the voice call are not broken by the change of network.
- Required components: mobile node with Mobile IP client, Home Agent, Foreign Agents / access gateways, the IP backbone, the EHR server and the specialist terminals, an authentication server (AAA) and IPsec gateways.

- Communication flow: (1) the ambulance enters a new network and discovers the FA; (2) it registers with the HA through the FA; (3) the monitor data go from the MN to the EHR server (directly or by reverse tunnel); (4) the physician's instructions go from the CN to the home address, are intercepted by the HA, tunnelled to the CoA and delivered to the paramedic; (5) when the ambulance moves again, steps 1 and 2 are repeated and the tunnel end is changed.

Student A – Performance and Requirement Analysis

Requirements of the traffic

Traffic	Rate	Delay requirement	Loss requirement
Voice (G.711)	64 kb/s (50 packets/s)	One-way ≤ 150 ms	≤ 1–2%
Vital signs and ECG	About 250 kb/s (100 packets/s)	≤ 500 ms	Very low (retransmission by TCP)
Video of the patient	About 1 Mb/s	≤ 400 ms	≤ 2%
Patient record access	Bursty	≤ 1 s	None (reliable transport)

Communication delay, packet delivery and interruption

Assumed one-way delays: radio access 30 ms, access network to backbone 15 ms, backbone MN-side to CN 20 ms, HA–FA path 25 ms, CN–HA 20 ms. Direct path (MN → CN or CN → MN with route optimisation): 30 + 15 + 20 = 65 ms. Through the home agent (triangular routing): CN → HA 20 ms + HA → FA 25 ms + FA → MN (15 + 30) ms = 90 ms, i.e. 25 ms more than the direct path; both values are within the 150 ms limit of voice and the 500 ms limit of vital signs, so the delay requirement is met even with the HA detour.

Interruption during movement: the ambulance travels at 60 km/h (16.7 m/s) through cells of 2 km, so it changes network about every 120 s. With a 1 s advertisement interval, the detection of the movement takes on average 0.5 s and the registration round trip 2 × (5 + 30) = 70 ms, so the interruption is about 0.57 s, and 57 packets of the vital-sign stream (100 packets per second) are lost per handoff (0.47% of the time lost: 0.57/120). With a 0.2 s advertisement interval and buffering the interruption falls to about 0.17 s and 17 packets, and the vital signs are retransmitted by TCP so that no data are lost. Packet delivery and connectivity: if the coverage is continuous, more than 99% of the time is connected; in areas without coverage (tunnels, damaged infrastructure) the connection is lost, and the data must be buffered until a new attachment exists.

Mobility requirements, security and reliability

Mobility requirements: seamless handoff between cellular and Wi-Fi, a handoff time below 0.5 s, support for movement at least 100 km/h and a fixed identity (home address). Security: mutual authentication of the MN and the HA, protection of the registration against replay and hijacking, encryption of the patient data (IPsec/TLS), access control and audit of the records. Reliability: redundant home agents, buffering and retransmission, re-registration with a short

lifetime, alternative access networks (multi-homing) and an ad hoc/vehicle-to-vehicle fallback when the infrastructure is lost.

Routing requirements created by mobility and evaluation of the suitability

Mobility requires that (1) the routes toward the mobile node are updated immediately after every move (binding update or tunnel change); (2) the routing is loop-free and uses fresh information; (3) the control overhead is small compared with the data; and (4) alternative routes exist if a link fails. If infrastructure disappears, a multi-hop ad hoc routing protocol (AODV) would be needed between ambulances.

Criterion	Mobile IP evaluation	Rating
Continuous connectivity and a fixed address	Yes: the home address is retained during movement	Good
Delay	65–90 ms one-way, within the limits	Good
Interruption	0.57 s with standard timers; 0.17 s with tuned timers	Acceptable after tuning
Overhead	20 bytes per packet (10% for 160-byte voice packets, 1.4% for 1,400-byte data)	Acceptable
Security	Authentication and IPsec available	Good
Infrastructure dependence	Needs the HA and access networks; not usable when the infrastructure fails	Limited – needs an ad hoc fallback

Conclusion of the evaluation: Mobile IP is suitable for ambulance communication in the presence of infrastructure, with fast handoff, route optimisation and security extensions; for areas without infrastructure it must be complemented by ad hoc routing.

ANSWER TO Q5 – Proactive Routing Protocols in Mobile Networks (6 Marks)

Shared Group Work

Proactive (table-driven) routing: every node keeps a route to every other node at all times by exchanging routing information periodically and when the topology changes; when a packet has to be sent, the route is already in the table, so there is no route discovery delay, but the control messages are sent continuously, whether there is traffic or not. The selected protocol is DSDV (Destination-Sequenced Distance-Vector), a distance-vector protocol made loop-free by destination sequence numbers.

Student A – Protocol Design

Principle of proactive routing and DSDV operation

DSDV: each node maintains a table with, for every destination, the next hop, the hop count (metric) and a sequence number created by the destination. Each node advertises its table to its neighbours periodically (full dump, for example every 15 s) and also immediately when something important changes (incremental or triggered update). The destination increases its own sequence number by 2 at every advertisement (even numbers); a node that detects a broken link to a neighbour sets the metric of the affected routes to infinity and increments their sequence number by 1 (odd numbers), so that an old route to the destination cannot be used again. Update rule (Section 17 of the standard): a received route replaces the existing one if it has a higher sequence number, or the same sequence number and a smaller hop count.

Sample topology and the routing information before and after a topology change

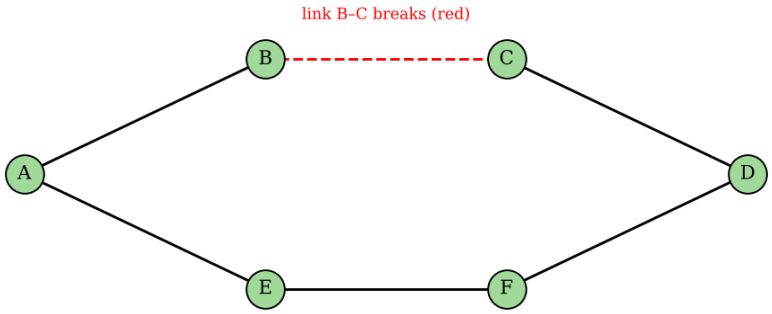


Figure 8: Sample topology A–F; the link B–C breaks (dashed red)

Topology: links A–B, B–C, C–D, D–F, F–E and E–A (a ring). Sequence numbers of the destinations before the change: A 102, B 118, C 234, D 312, E 150, F 276. Routing table of node A (before the change):

Destination	Next hop	Hop count	Sequence number
B	B	1	118
C	B	2	234

Destination	Next hop	Hop count	Sequence number
D	B	3	312
E	E	1	150
F	E	2	276

Change: node B moves away and the link B–C breaks. Node B detects it (no reply to hello/missing advertisements), marks C and D (reached through C) as unreachable (metric ∞) and increases their sequence numbers by one (C 235, D 313), and sends a triggered update. Node A receives this update from the next hop B and also sets C and D to ∞:

Destination	Next hop	Hop count	Sequence number	Comment
B	B	1	118	unchanged
C	B	∞	235	broken (odd number)
D	B	∞	313	broken (odd number)
E	E	1	150	unchanged
F	E	2	276	unchanged

Recovery: C and D hear that they are unreachable (or their next periodic advertisement comes) and issue new even sequence numbers (C 236, D 314); the updates travel D → F → E → A, and node A accepts them because the sequence numbers are higher than 235 and 313:

Destination	Next hop	Hop count	Sequence number
B	B	1	118
C	E	4	236
D	E	3	314
E	E	1	150
F	E	2	276

The route to D changes from A → B → C → D (3 hops) to A → E → F → D (3 hops) and the route to C from 2 hops to 4 hops (A → E → F → D → C). The sequence numbers guarantee that the stale route (through B) is never accepted again, so no routing loop is created.

ALGORITHM DSDVLinkBreakHandling(Node, LostNeighbor)
INPUT:
Node – node that detects the broken link; LostNeighbor – neighbour that is no longer reachable
OUTPUT:
Updated RouteTable and a triggered update message
BEGIN

```
ChangedSet ← EMPTY_SET
FOR each Destination in RouteTable DO
IF RouteTable[Destination].NextHop = LostNeighbor AND
RouteTable[Destination].HopCount < INFINITY THEN
RouteTable[Destination].HopCount ← INFINITY
RouteTable[Destination].SequenceNumber ← RouteTable[Destination].SequenceNumber +
1
ADD Destination TO ChangedSet
END IF
END FOR
IF ChangedSet is not empty THEN
SEND TriggeredUpdate(ChangedSet entries) TO all neighbors
END IF
RETURN RouteTable
END
```

Student B – Performance Investigation

Routing-control overhead, route availability and control-message transmission

Overhead analysis: with N nodes and a full-dump period P, every node sends one update of N entries every P seconds; the number of update transmissions per second is N/P (for N = 30 and P = 15 s, 2 per second) and the size of each update grows with N, so the periodic overhead in bytes grows as N^2/P. Triggered updates are added for every link change, which grows with the mobility. Simulation (Q7 describes the simulator): 1,000 m × 1,000 m, radio range 250 m, random waypoint movement with a maximum speed of 10 m/s, 10 flows of 4 packets/s, 200 s per run (first 30 s ignored), mean of 3 runs.

Number of nodes	Control transmissions per second	Control bytes per second	Per node (transmissions/s)	Packet delivery ratio	Connectivity (upper limit)	Route availability (PDR/connectivity)	Ave dela (ms
10	7.2	340	0.72	0.504	0.540	0.93	18.4
20	70.3	3830	3.52	0.780	0.870	0.90	23.3
30	181.9	11935	6.06	0.831	0.915	0.91	22.5
50	448.9	44000	8.98	0.849	0.977	0.87	22.4

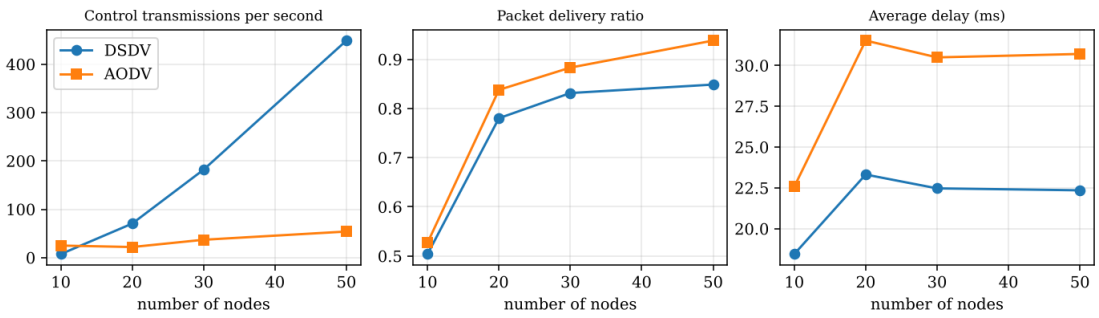


Figure 9: DSDV and AODV for 10 to 50 nodes (maximum speed 10 m/s)

Convergence and route availability: a topological change is signalled at once by a triggered update, but the new route becomes available only when the destination issues a higher sequence number (up to one periodic interval, 15 s, in the worst case, and the update has to travel hop by hop); during this time the packets that use the stale route are lost. The route availability, i.e. the fraction of the connected time in which a valid route was available, was 0.93, 0.90, 0.91 and 0.87 for 10, 20, 30 and 50 nodes. Network size: the control overhead grows much faster than linearly: from 7.2 transmissions/s at 10 nodes to 448.9 at 50 nodes (62 times, for 5 times more nodes), which is 0.72 and 8.98 transmissions per second per node. Delay and delivery: because the route is always ready, the delay is only the forwarding delay (about 18 to 23 ms, 10 ms per hop) with no waiting for a route discovery; the delivery ratio was 0.50, 0.78, 0.83 and 0.85, close to the connectivity limit of the topology (0.54, 0.87, 0.92 and 0.98) and decreasing in dense networks because stale entries are present until they are updated.

Conclusion: DSDV gives the lowest delay and immediate availability of routes, but its control overhead rises steeply with the number of nodes and the mobility, and it does not scale to large or very dynamic networks.

ANSWER TO Q6 – Reactive Routing and Protocol Comparison (7 Marks)

Student A – Reactive Routing Investigation

Concept: a reactive (on-demand) protocol does not maintain routes to all destinations. A route is searched only when a source has a packet to send and has no valid route; the route is then maintained only as long as it is used. Control overhead is therefore generated only when needed, at the cost of a route discovery delay for the first packet. AODV (Ad hoc On-demand Distance Vector) is the reference protocol; it uses hop-by-hop routing tables and destination sequence numbers to be loop-free.

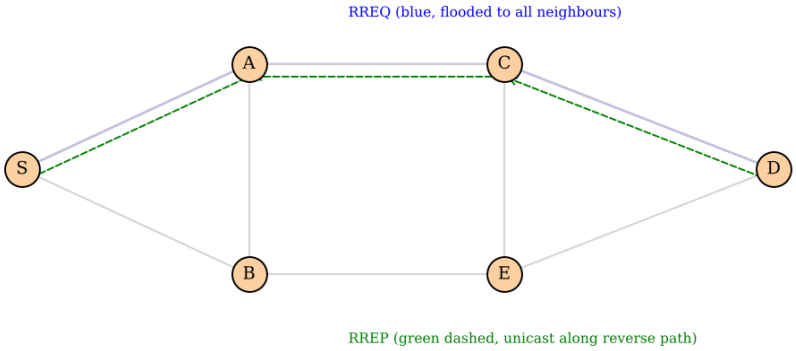


Figure 10: AODV route discovery from S to D (RREQ flooded, RREP returned along the reverse path)

RREQ, RREP and RERR

- Route Request (RREQ): broadcast by the source; contains the source address and sequence number, the broadcast ID, the destination address and the last known destination sequence number, and the hop count. Each receiving node discards a request that it has already processed (same source and broadcast ID), creates a reverse route towards the source, and rebroadcasts the request, unless it is the destination or has a fresh route to it.
- Route Reply (RREP): created by the destination (or an intermediate node with a fresh enough route) and unicast back along the reverse path; each node on the path installs the forward route (next hop = the node from which the reply arrived, hop count, destination sequence number, lifetime).
- Route Error (RERR): sent when a node detects that the next hop of an active route is no longer reachable; it lists the unreachable destinations with increased sequence numbers and is sent to the upstream neighbours (the precursors) up to the source, which removes the routes and, if it still has data, starts a new route discovery.

Route discovery and route maintenance – desk check of Figure 10

Step	Node	Event	Decision	Route state
1	S	Has data for D, no route	Create RREQ(S, D, id 1)	UNKNOWN
2	S	Broadcast RREQ	Neighbours A and B receive it	DISCOVERY

Step	Node	Event	Decision	Route state
3	A, B	Receive RREQ	Record reverse route to S; rebroadcast	DISCOVERY
4	C, E	Receive RREQ from A / B	Record reverse route; rebroadcast; a duplicate from the other branch is discarded	DISCOVERY
5	D	Receives RREQ via C (hop count 3)	Destination reached: create RREP	VALID
6	C, A	Receive RREP	Install forward route to D; relay to S	VALID
7	S	Receives RREP	Install route S → A → C → D (3 hops)	VALID
8	S	Data packet	Forward to A	ACTIVE

```
ALGORITHM ProcessRouteError(Node, RERR)
INPUT:
Node – node receiving the route error; RERR = {list of (Destination, SequenceNumber)}
OUTPUT:
Updated RouteTable and propagated error or new discovery
BEGIN
AffectedSet ← EMPTY_SET
FOR each (Destination, Seq) in RERR DO
IF RouteTable[Destination].NextHop = RERR.Sender THEN
RouteTable[Destination].State ← INVALID
RouteTable[Destination].SequenceNumber ← Seq
ADD Destination TO AffectedSet
END IF
END FOR
IF AffectedSet is not empty THEN
FOR each Neighbor in Precursors(AffectedSet) DO
SEND RouteError(AffectedSet) TO Neighbor
END FOR
IF Node has own data for a Destination in AffectedSet THEN
RouteRequestID ← RouteRequestID + 1
BROADCAST RouteRequest(Node, Destination, RouteRequestID)
END IF
END IF
RETURN RouteTable
END
```

Effect of mobility on route discovery

Each time a link of an active route breaks, an RERR is generated and a new discovery (flooding of about N transmissions) is needed; therefore the number of discoveries and the control traffic grow with the mobility. Simulation with 30 nodes (maximum speed in m/s):

Maximum speed	Route discoveries in the run	Control transmissions per second	Control bytes per second	Packet delivery ratio	Average delay (ms)	Average discovery delay (ms)
1.5 m/s	70	4.5	231	0.960	34.4	115
5 m/s	134	25.5	1320	0.922	30.5	109
10 m/s	208	36.9	1908	0.883	30.5	108
20 m/s	300	49.1	2532	0.890	34.2	106

When the speed increases from 1.5 to 20 m/s the number of discoveries grows from 70 to 300 and the control transmissions from 4.5 to 49 per second (11 times), but the overhead stays well below that of DSDV and the delivery ratio remains between 0.89 and 0.96. The first packet of every discovery waits about 100 to 115 ms (2 × hops × 10 ms for the RREQ/RREP exchange plus processing).

Student B – Comparative Verification (AODV and DSDV)

Aspect	DSDV (proactive)	AODV (reactive)
Route discovery mechanism	None: routes to all nodes are kept by periodic and triggered table exchange	On demand: RREQ flooding and RREP unicast when a route is needed
Routing-table requirement	Entry for every destination in the network (N entries per node)	Entries only for the active destinations and neighbours (reverse and forward routes with lifetimes)
Control overhead	Periodic + triggered, independent of the traffic; grows quickly with N and mobility (30 nodes, 10 m/s: 181.9 tx/s)	Demand-driven; depends on the traffic and the route breaks (36.9 tx/s for the same case, 4.9 times less)
Initial communication delay	None (the route exists): 22.5 ms average	Route discovery delay of about 108 ms for the first packet; average delay 30.5 ms
Mobility handling	Triggered updates, sequence numbers; a new route may wait for a destination update, so stale routes cause loss	RERR and new discovery only for the routes in use; a break affects only active flows

Aspect	DSDV (proactive)	AODV (reactive)
Packet delivery (30 nodes, 10 m/s)	PDR 0.831	PDR 0.883
Suitable scenarios	Small, fairly stable networks with frequent traffic and very low delay requirements	Large or highly mobile networks with intermittent traffic and limited bandwidth

Selection of the protocol for a given network condition: using the algorithm SelectRoutingApproach (Section 22 of the standard): if DelayRequirement = VERY_LOW and TrafficRate = HIGH, PROACTIVE is selected; if MobilityRate = HIGH and TrafficRate = INTERMITTENT, or if the bandwidth is limited and the traffic is low, REACTIVE is selected. For a small static office-type mesh with continuous traffic DSDV is a good choice; for a vehicle or rescue-team network AODV is better.

Note on the small network (10 nodes): in the simulation AODV needed more control traffic than DSDV (24.9 against 7.2 tx/s) because the 10 nodes are often partitioned (connectivity 0.54) and the sources repeat the discovery of unreachable destinations; AODV wins when the network is larger and the routes needed are fewer than all the possible routes.

Joint Question

For a highly dynamic mobile network with frequent topology changes, would you select DSDV or AODV? Decision: AODV. Justification: (1) Routing overhead: in the simulation with 30 nodes at a maximum speed of 20 m/s, DSDV needed 238.9 control transmissions per second (17591 bytes/s) and AODV only 49.1 (2532 bytes/s), 4.9 times less; with 50 nodes at 10 m/s the factor is 8.3 (448.9 against 53.9 tx/s). (2) Delivery: AODV delivered 89.0% of the packets against 79.9% for DSDV, because DSDV tables are often stale after link changes. (3) Delay: the price of AODV is the discovery delay of about 106 ms for the first packet of a route and a slightly higher average delay (34.2 ms against 22.7 ms), which is acceptable for most emergency services. (4) Mobility: AODV repairs only the routes in use, while DSDV has to keep every route fresh for all the nodes, and every link change generates network-wide updates. DSDV would be chosen only if the delay requirement is extremely low and the topology is stable.

ANSWER TO Q7 – Design a Mobile Emergency Communication Network (10 Marks)

The scenario is solved collaboratively; the independent proposals of Stage 1 are retained exactly as they were first written, followed by the comparison, the joint final design, the validation and the defence.

Stage 1 – Independent Design Proposal

Student A (retained initial proposal)

- Wireless network architecture: a hybrid network: the rescue vehicles carry on-board units with Wi-Fi/ad hoc radio and a cellular/satellite uplink; where the cellular network still works, vehicles attach to it; otherwise they form a multi-hop mobile network around a command vehicle that acts as gateway.
- Mobility-management strategy: each node keeps a permanent home address; location is tracked by registration with a home agent in the control centre; handoff between access networks uses signal strength with a 5 dB hysteresis.
- Role of Mobile IP: each vehicle and team is a Mobile Node with a home address in the control centre network, the Home Agent in the control centre tunnels packets to the current care-of address, and the gateway vehicle acts as the Foreign Agent.
- Communication model: voice and short messages from the field to the control centre and between the teams (many-to-one and peer-to-peer), periodic position reports and priority for the critical messages.

Student B (retained initial proposal)

- Routing strategy: a multi-hop ad hoc routing protocol on all the mobile nodes, because the topology changes constantly and fixed infrastructure may be missing.
- Proactive/reactive selection: reactive (AODV), because the traffic is intermittent (reports and short messages), the topology is highly dynamic and the bandwidth is limited.
- Route maintenance: link-layer feedback and hello messages detect the failure of a link; the node sends an RERR to the sources; the sources start a new route discovery; routes expire after 10 s without use; local repair is tried first.
- Failure-handling mechanism: alternative routes through other nodes, retransmission of the critical messages, store-and-forward buffering when the destination is unreachable and a timeout to report the disconnection to the user.
- Performance parameters: end-to-end delay (target $\leq$ 150 ms for voice, $\leq$ 1 s for messages), packet delivery ratio ($\geq$ 90%), routing overhead (control packets and bytes per second), route discovery delay and connectivity.

Stage 2 – Compare the Two Designs

Item	Student A design	Student B design	Comparison / decision
Agreement	Multi-hop wireless network without dependence on fixed infrastructure	Same	Both reject a pure cellular design
Difference	Mobility management and Mobile IP at the infrastructure level	Routing between mobile nodes	The designs are complementary: A is the global addressing, B is the local routing
Strengths	Fixed identity, Internet/hospital access through the gateway	Fast adaptation to topology change, low overhead	Combine both
Weaknesses	No routing protocol specified; HA at the control centre is a single point of failure and may be unreachable	No global addressing or connection to the Internet / hospital	Add routing to A and Mobile IP to B
Mobile IP requirement	Required for all nodes	Not considered	Mobile IP only for the sessions that need a fixed address and Internet access through the gateway
Routing suitability	Not stated	AODV is suitable for the intermittent, dynamic traffic	Select AODV
Expected overhead	Registration per move (2 messages) and 20 bytes per tunnelled packet	Discovery flood of about N packets per new route	Small; see validation
Expected delay	Tunnel adds up to 25 ms; registration up to 190 ms	Discovery adds about 100 ms to the first packet; 10 ms per hop	Acceptable for emergency voice and messages
Node mobility	Registration and handoff	Route repair	Both are needed

Stage 3 – Joint Final Design

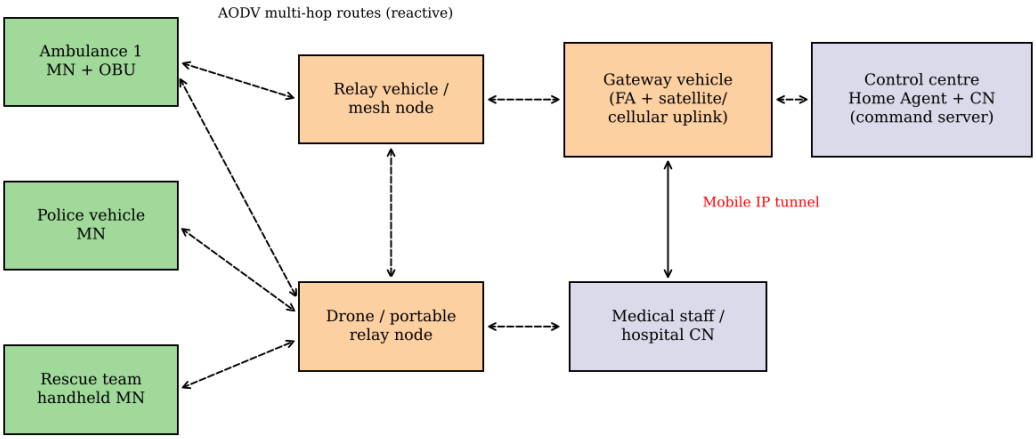


Figure 11: Final mobile emergency communication network

Network architecture

Mobile nodes: ambulances, police vehicles, rescue-team handhelds and medical personnel devices; relay nodes: relay vehicles and portable (for example drone-mounted) relay nodes; gateway vehicle with satellite/cellular uplink to the control centre (Home Agent and correspondent node) and hospitals. The wireless links between all mobile nodes use an ad hoc mode, forming a multi-hop mesh; the routing infrastructure is made of the routing function in every mobile node plus the gateway.

Mobile IP components, registration and tunnelling

- Mobile Node: every vehicle and team terminal, with a permanent home address in the control-centre network (for example 10.10.0.x).
- Home Agent: in the control centre (a redundant pair), keeps the bindings and authenticates the registrations.
- Foreign Agent: the gateway vehicle and any node that has an uplink, advertising a care-of address to the ad hoc network.
- Correspondent Node: the command server, hospital systems and other teams outside the ad hoc area.
- Registration: the MN receives the advertisement of the gateway (through AODV routes), sends a registration request (HoA, CoA, lifetime 60 s, identification, HMAC authenticator) to the HA through the FA, and the HA returns the reply; the MN re-registers when it changes the gateway or before the lifetime expires.
- Tunnelling: the packets for the HoA reach the HA, are encapsulated (IP-in-IP, 20 bytes) towards the CoA of the gateway and delivered to the MN over the AODV multi-hop route; the packets between two nodes inside the ad hoc area do not use Mobile IP but are routed directly by AODV (no triangular routing).

Selected routing protocol and justification

AODV (reactive), because the topology is highly dynamic (vehicles at up to 20 m/s), the traffic is intermittent, bandwidth is limited and AODV has the lowest overhead among the options (Q6: 49 against 239 control transmissions per second at 20 m/s with 30 nodes) and the best delivery (0.89 against 0.80); the 100 ms discovery delay is acceptable, and the critical teams can keep a permanently active route (periodic hello and no route expiry for the command link).

Failure handling

Failure	Detection	Handling
Node failure	Missing hello messages / link-layer feedback (no acknowledgement)	Mark the links invalid, RERR to the sources, new discovery avoiding the failed node; critical messages retransmitted
Link failure (node out of range)	Same	Local repair at the node that detects the break; otherwise the source rediscovers the route
Network disconnection (partition)	Route discovery fails after the retries (3 RREQ), timer of 5 s	Store-and-forward buffering of the critical messages, user notification, relay drone/vehicle repositions to restore the link, periodic retry; alternative gateway
Gateway/uplink failure	Registration reply missing; advertisement lost	Select another gateway vehicle and re-register (the HA updates the binding); local communication continues without Mobile IP
HA failure	No registration reply	Standby home agent in the control centre

ALGORITHM EmergencyRouteMaintenance(Node, Route, Message)
INPUT:
Node – node that forwards the message; Route – active route; Message – packet with Priority
OUTPUT:
Delivery status
BEGIN
IF LinkStatus(Route.NextHop) = UP THEN
FORWARD Message TO Route.NextHop
RETURN FORWARDED
END IF
Mark Route INVALID
IF LocalRepairPossible AND HopCount(Route to destination) ≤ 2 THEN
BROADCAST RouteRequest with limited TTL
IF RouteReply received before REPAIR_TIMEOUT THEN
FORWARD Message TO new NextHop

```
RETURN REPAIRED
END IF
END IF
SEND RouteError toward Message.Source
IF Message.Priority = CRITICAL THEN
ADD Message TO StoreBuffer
START TIMER bufferTimer
RETURN BUFFERED
ELSE
DROP Message
RETURN DROPPED
END IF
END
```

Performance evaluation

The evaluation uses delay, packet delivery and routing overhead; the analytical (Student A) and experimental (Student B) validations are given in Stage 4.

Stage 4 – Divide the Validation Responsibilities

Student A – Analytical Validation

- Mobile IP overhead: the tunnel adds 20 bytes per packet: 10% on a 160-byte voice packet (with 40 bytes of headers), 1.4% on a 1,400-byte packet; the signalling is 2 messages (about 100 bytes) per move: for a vehicle that changes the gateway every 60 s this is about 1.7 bytes/s, negligible. With 100 mobile nodes the HA receives about $100/60 \approx 1.7$ registrations per second.
- Mobility-related delay: detection and registration with a 0.2 s advertisement interval: 100 ms average detection plus a registration round trip through the multi-hop path (3 hops × 2 × 10 ms + 40 ms to the HA = about 100 ms), about 200 ms per move; the tunnelled path adds up to 25 ms.
- Packet delivery: with 50 packets/s voice, each move loses about 10 packets (0.2 s); a node that moves every 60 s loses 0.33% of the packets from the movement alone; adding the loss of the route breaks (about 10 to 11% in the simulation at 10 m/s) gives an expected delivery of about 88–89% for AODV at 30 nodes, matching the measurement.
- Communication interruption: about 0.2 s per gateway change and about 0.1 s per route discovery; in total less than 1% of the time for typical movement.
- Justification of the mobility management: Mobile IP is used only for the sessions to the outside (control centre, hospitals), where a fixed address is needed, while internal communication uses direct AODV routing, which avoids the triangular route and keeps the overhead small.

Student B – Experimental Validation

Test environment: a discrete-time simulator written in Python for this assignment (0.5 s steps). Area 1,000 m × 1,000 m; unit-disk radio with range 250 m; random waypoint mobility (speed uniformly between 20% and 100% of the maximum, no pause); 10 constant-bit-rate flows of 4 packets/s between random pairs; 10 ms delay per hop; 200 s per run, the first 30 s ignored; results are the mean of 3 random seeds. DSDV: full dump every 15 s, triggered updates on link change (propagating up to 5 hops per step), sequence numbers and the odd/even rule. AODV: RREQ flooding with duplicate suppression, RREP along the reverse path, RERR and rediscovery on link break, no hello messages (link-layer feedback). Limitations: no MAC-layer collisions or interference and ideal link-break detection, so the real overhead of both protocols, especially the broadcast floods of AODV, would be higher.

Test	Configuration	Purpose
T1	10 mobile nodes, maximum speed 10 m/s	Small network (partitions)
T2	20 mobile nodes, 10 m/s	Medium network
T3	30 mobile nodes, 10 m/s	Typical emergency group
T4	50 mobile nodes, 10 m/s	Large deployment
T5	30 nodes, 1.5 m/s	Rescue teams on foot
T6	30 nodes, 5 m/s	Mixed teams
T7	30 nodes, 20 m/s	Fast vehicles (72 km/h)

Results for different numbers of mobile nodes (maximum speed 10 m/s):

Nodes	Protocol	PDR	Average delay (ms)	Control tx/s	Control bytes/s	Connectivity
10	DSDV	0.504	18.4	7.2	340	0.540
10	AODV	0.527	22.6	24.9	1292	0.540
20	DSDV	0.780	23.3	70.3	3830	0.870
20	AODV	0.838	31.5	21.9	1126	0.870
30	DSDV	0.831	22.5	181.9	11935	0.915
30	AODV	0.883	30.5	36.9	1908	0.915
50	DSDV	0.849	22.4	448.9	44000	0.977
50	AODV	0.938	30.7	53.9	2791	0.977

Results for different mobility conditions (30 nodes):

Max speed	Protocol	PDR	Average delay (ms)	Control tx/s	Control bytes/s	Connectivity
1.5 m/s	DSDV	0.946	28.1	71.8	3607	0.966

Max speed	Protocol	PDR	Average delay (ms)	Control tx/s	Control bytes/s	Connectivity
1.5 m/s	AODV	0.960	34.4	4.5	231	0.966
5 m/s	DSDV	0.881	23.9	130.1	7655	0.940
5 m/s	AODV	0.922	30.5	25.5	1320	0.940
10 m/s	DSDV	0.831	22.5	181.9	11935	0.915
10 m/s	AODV	0.883	30.5	36.9	1908	0.915
20 m/s	DSDV	0.799	22.7	238.9	17591	0.957
20 m/s	AODV	0.890	34.2	49.1	2532	0.957

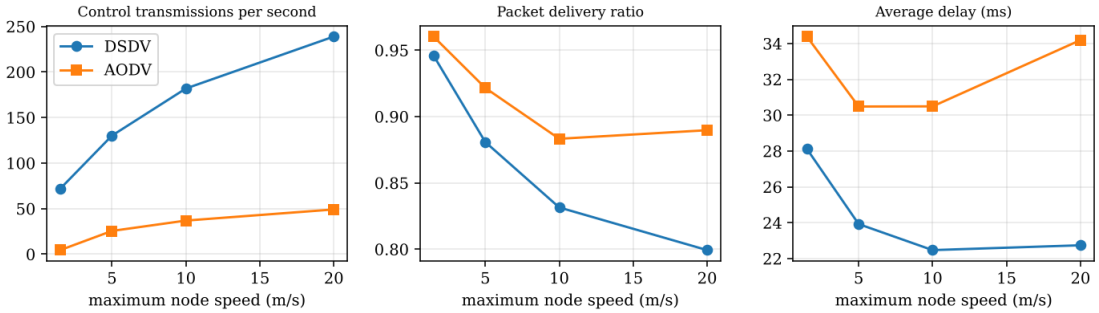


Figure 12: DSDV and AODV at different speeds (30 nodes)

Experimental evidence: (1) proactive versus reactive: DSDV has a lower average delay (about 22 ms against 30 to 34 ms) but a control overhead that is 1.5 to 8 times higher for 20 or more nodes and rises with both the size and the speed; AODV's discovery delay was 93 to 115 ms. (2) Delivery: AODV delivered 84 to 96% of the packets against 50 to 95% for DSDV; the difference is the largest at high mobility (at 20 m/s: 0.890 against 0.799). (3) Both protocols are limited by the connectivity of the topology (for 10 nodes only 54% of the flow time had a path). (4) For the emergency network (20 to 50 nodes, 5 to 20 m/s) AODV is therefore the better choice.

Stage 5 – Cross-Defence

Student A – why was the selected routing protocol (AODV) appropriate? The emergency network has moving vehicles and teams, intermittent traffic (reports and messages), limited bandwidth and no stable topology. AODV creates routes only when needed and repairs only the routes in use, so its overhead is low (49 against 239 control packets per second at 20 m/s) and it delivers more packets (0.89 against 0.80). It supports multi-hop routing without any infrastructure, uses sequence numbers to avoid loops and detects failures quickly. Its drawbacks, the discovery delay of about 100 ms and the flooding overhead, are acceptable for emergency voice and message traffic.

Student B – how does Mobile IP maintain communication when a mobile node changes its network location? The MN keeps its home address; at the new location it discovers the agent, obtains a care-of address and registers it with the Home Agent, which updates the binding HoA

↔ CoA. The Correspondent Node continues to send to the home address; the HA intercepts the packets, encapsulates them to the new CoA and the FA delivers them to the MN. Since the home address does not change, the transport connection (TCP) and applications continue; only a short interruption (about 0.2 s with the tuned timers) occurs during the registration.

Stage 6 – Joint Evaluation

- Why the architecture is suitable: it does not depend on fixed infrastructure, is deployed immediately, extends the coverage by multi-hop relays, tolerates node failures and still connects to the control centre and the hospitals through the gateway.
- Why Mobile IP is required: for the sessions with the control centre and hospitals, where the vehicles must keep a fixed address while the gateway or the access network changes; it is not required for local communication between nodes inside the ad hoc network, which uses AODV only.
- Why AODV is appropriate: low overhead with dynamic topology and intermittent traffic, best delivery in the experiments at 20 to 50 nodes and speeds of 5 to 20 m/s.
- Network overhead introduced: AODV control traffic of about 22 to 54 transmissions/s (1.1 to 2.8 kB/s) for 20 to 50 nodes at 10 m/s; Mobile IP: 20 bytes per tunnelled packet and about 100 bytes of signalling per move.
- How mobility affects performance: more moves mean more route breaks, rediscoveries (70 to 300 per run when the speed rises from 1.5 to 20 m/s) and packet losses; the delivery ratio falls from 0.96 to 0.89 and the control traffic rises 11 times.
- What happens when connectivity is lost: route discovery fails, the sender buffers the critical packets and retries, the user is informed, relay nodes can be repositioned, and, for the Mobile IP sessions, the node re-registers with another gateway as soon as it hears one.
- Conditions in which the solution may not be appropriate: very large networks (hundreds of nodes, flooding becomes expensive), continuous high-rate or very delay-sensitive traffic (a proactive protocol or infrastructure would be better), very sparse networks with long partitions (a delay-tolerant approach is needed), severe interference or security threats without key management, and the single dependence on one gateway.

Conclusion: What characteristics should a network designer analyse before selecting Mobile IP and a routing protocol for a highly mobile wireless environment? The designer should analyse: (1) the availability and reliability of the infrastructure and the need for a fixed address or Internet access (decides whether Mobile IP is needed); (2) the mobility pattern and speed (handoff and link-break rate); (3) the network size and density (overhead scaling); (4) the traffic pattern (continuous or intermittent) and the delay and loss requirements (proactive or reactive); (5) the available bandwidth and the energy of the nodes; (6) the connectivity and the partition risk; (7) the security and the failure-handling needs; and (8) the cost of the overhead, which has to be verified by analysis and by simulation or experiment before the decision.

Answers to the Verification (VIVA) Questions

- If a mobile node moves from one network to another during an active session, what happens to its IP connectivity and how does Mobile IP address the problem? Without Mobile IP the node gets a new address in the new network, and the connections that use the old address are broken. With Mobile IP the node keeps its home address, registers the new care-of address with the Home Agent (590 ms in the worked example, about 190 ms with tuned timers), and the HA tunnels the packets to the new location, so the session continues after a short interruption.
- If the number of mobile nodes increases significantly, how would AODV and DSDV differ in routing overhead? DSDV: every node advertises a table with N entries, so the overhead grows roughly as N^2 (7.2 tx/s at 10 nodes to 448.9 tx/s at 50 nodes in the simulation), even without traffic; AODV: the overhead depends on the number of routes in use and their breaks, and grew from 24.9 to 53.9 tx/s in the same range; for large networks AODV is much cheaper, but each discovery floods the network.
- If the network topology changes very frequently, which routing approach would you select and why? A reactive protocol (AODV): it repairs only the routes in use and avoids network-wide table updates; in the simulation at 20 m/s it needed 49 control transmissions/s against 239 for DSDV and delivered 89% of the packets against 80%.
- If the fixed infrastructure becomes unavailable during a disaster, how would an ad hoc network support communication? The mobile nodes form a multi-hop network by themselves: each node forwards packets for the others, a routing protocol finds a path, and when a node or link fails another path is used; a gateway vehicle (satellite or cellular uplink) can reconnect the network to the control centre and hospitals when available.

End of the answers.